

The Spine — the chain in one pass

One continuous explanation of the aftermarket for the CFM56 (the engine that powers the 737NG and A320ceo) and FTAI Aviation's place in it, at the level of mechanism. Descriptive throughout; every figure is a binding figure with its source; where two figures compete, both are given. Takes no view.

Contents

The chain in one pass.....	2
How to read the Spine	3
S1. The machine: a turbofan, its modules, the life limits that force it open, and the margin that erodes	3
S2. What a shop visit is and costs, and why the parts bill is where the money sits.....	7
S3. Where the money sits across the chain: a first margin map, bases stated, nothing ranked	11
S4. Who owns engines and how they pay: leases, reserves, engine values, green time	14
S5. Why 2026 looks the way it does: the fleet that only shrinks, retirements that did not come, new engines that did not relieve, airframers that did not deliver.....	19
S6. The translation table in one page: each driver, the requirement it creates, what it does not stress.....	23
S7. The shop-visit market and the OEM's pricing: demand plateau, capacity shortage, list-price escalation, the OEM's own offerings	26
S8. The substitutes for new parts: used material from teardowns, and PMA and DER	30
S9. FTAI's answer: the Module Factory and the Aerospace Products segment.....	35
S10. FTAI's capital: Aviation Leasing, the Strategic Capital vehicles, the money that flows between them, and the 2026 securitization	40
S11. FTAI Power: a second life for cores, the J&F order, the market it sells into.....	45
S12. The company, its numbers and what remains unknown	48
Tensions carried in the Spine	52
Unknowns carried in the Spine	53
Sources.....	53

THE SPINE

The chain in one pass

This is the whole chain, start to finish, in one reading: the engine, what it costs to keep flying, who pays, why the fleet that powers the world's older narrowbodies is still flying in 2026, how that fact became a shop-visit plateau and a parts shortage, who sells the substitutes for new parts, and how FTAI Aviation built a module business, a capital-management business and a power business on top of it. Each section ends with a pointer to the Part that carries the depth. A reader who finishes it should be able to derive the system's main consequences without having read the Parts.

HOW TO READ THE SPINE

The Spine follows the flow of the industry, not the order of the Parts. It starts with a machine and ends with a company's balance sheet. It defines, in a short box at first use, only the terms a reader cannot do without; every other term is defined in the Part that owns it. It carries only the numbers another link depends on, and it presents each of them the way the research fixed them: with a source, a date, and, where two sources disagree, both figures and no average. Ten places in the chain carry a disagreement that bears directly on the economics; the Spine presents each where the chain reaches it and lists them at the end. Nothing here is ranked and nothing is recommended.

S1. The machine: a turbofan, its modules, the life limits that force it open, and the margin that erodes

Everything in this primer is a consequence of four physical facts about one engine. The engine is built in separable sections. Inside those sections sit parts that are scrapped on a clock. The hot section loses temperature margin as it wears. And the engine is only as usable as its paperwork. This section states the four facts and the arithmetic that connects them.

TURBOFAN: FAN, BYPASS AND CORE

A turbofan is a jet engine with a large fan on the front. The fan pushes most of the air around the hot middle of the engine, not through it; the air that goes around is the bypass stream. Only a small share of the air goes through the middle, where fuel is burned and turbines are driven. The turbines spin the fan, and the fan does most of the pushing. The CFM56 is a high-bypass turbofan with two rotating assemblies, one inside the other, connected only by the air flowing between them. [Part I §1]

CFM56 (-7B AND -5B), THE 737NG AND A320CEO IT POWERS, AND THE V2500 (THE A320CEO'S ALTERNATIVE ENGINE)

The CFM56 is a family of turbofans of which more than 35,000 have been delivered. Published counts of engines "in service" are about 23,000 (CFM website, undated), about 24,000 as of September 2025 (Aviation Business News) and more than 22,800 at end-2025 (Safran), and none states whether spare and stored engines are included. Two variants matter here. The -7B is the only engine ever fitted to the Boeing 737NG (Next Generation), the 737-600 to -900ER built from about 1997 to 2019. The -5B powers about six in ten Airbus A320ceo (current engine option) aircraft, built from about 1988 to 2020; the other four in ten carry the V2500, the alternative engine from the IAE consortium. The last -7B for a commercial 737NG was delivered by May 2019 and installed-engine production of the -5B ended in 2022. No new CFM56 is built for an airframe, so every CFM56 that flies tomorrow exists today. [CFM website, undated; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; GE Aerospace, 2019; ISTAT (the International Society of Transport Aircraft Trading), undated]

ORIGINAL EQUIPMENT MANUFACTURER (OEM), AND CFM INTERNATIONAL

The original equipment manufacturer (OEM) is the company that holds the regulator's approval of the engine's design and built it. It is the sole source of new parts at its published catalogue price, it controls the manual that says how the engine may be repaired, and it runs its own repair shops and service contracts. For the CFM56 the OEM is CFM International, the joint venture of GE Aerospace and Safran Aircraft Engines, described everywhere as 50/50 and carried here as that widely published figure; GE builds the core and Safran the fan, booster, low-pressure turbine and gearbox. The same venture makes the LEAP (CFM's new-generation engine), which replaces the CFM56 on the A320neo and the 737 MAX (the re-engined 737). [CFM website, undated; Part I §2; Part IV §25]

The engine comes apart at a few large bolted joints, and the sections between the joints are the units in which this whole market trades.

MODULE, AND THE CORE AS THE HOT MODULE

A module is a section of the engine designed to be removed and replaced as a unit. It is bolted to its neighbours at flanges and carries its own part number, serial number and maintenance record, so it can move from one engine to another when its build standard, records and part numbers allow. The CFM56 has three gas-path major modules, the fan and booster, the core and the low-pressure turbine (LPT), plus an accessory gearbox outside the gas path. The core, also called the gas generator, is the hot, fast, expensive middle: the high-pressure compressor (HPC), the combustor and the high-pressure turbine (HPT). Sources count the modules as three, four, "four to five" or six, depending on the level counted and whether the gearbox is included; FTAI's own releases convert at three modules per engine ("450 modules (150 engines)", 26 Feb 2025). In FTAI's exchange business "core" has a second meaning, the unserviceable unit a customer hands back; the Spine says "returned core" for that sense. [Aircraft Commerce Issue 58, 2008; FTAI release, 26 Feb 2025; Part I §3; reconciliation R-035]

EXHIBIT S.1 — THE CFM56-7B MODULES, WHAT EACH DOES, AND THE LIFE-LIMIT CLOCK INSIDE EACH

Major module	Contents	What it does	Life-limited parts and certified limit (engine flight cycles)
Fan and booster	61-inch fan with 22 blades; three-stage booster; fan frame	Moves the bypass air that makes most of the thrust; pre-compresses the core air	Fan disk and booster spool, 30,000
Core (HPC, combustor, HPT)	Nine-stage compressor; combustor; single-stage turbine	Compresses, burns, and extracts the power that drives the compressor; the hottest and costliest section to restore	HPC spools and disk, HPT disk, all 20,000
Low-pressure turbine	Four-stage turbine and rear frame	Drives the fan and booster through the long inner shaft	LPT stage 1 disk, 25,000
Accessory gearbox	Gear train driven off the core	Powers the fuel, oil and hydraulic pumps and the generator	None in the records packs the research reached

Source: Aircraft Commerce, CFM56-7B Owner's & Operator's Guide, Issue 58, 2008; AJW Group (an engine-parts company) mini-pack for engine serial number (ESN) 889979, 2022; StandardAero mini-pack for ESN 892820, Mar 2025; CFM design policy per Aircraft Commerce Issue 34, 2004. Compiled from Part I Exhibits 1.3 and 1.5. Accessed 2026-10-03.

LIFE-LIMITED PART (LLP) AND THE ENGINE FLIGHT CYCLE (EFC)

An engine flight cycle (EFC) is one take-off and landing; a mature short-haul narrowbody flies roughly 1,000 to 1,500 cycles a year (Aircraft Value News). A life-limited part (LLP) is a disk, spool or shaft whose failure the engine case cannot be designed to contain. It is certified for a fixed number of cycles and must be scrapped at that limit whatever its condition. On the -7B the limits are 30,000 cycles in the fan and booster, 20,000 in the compressor and high-pressure turbine, and 25,000 in the low-pressure turbine. The parts are the innermost parts of the engine, so reaching a limit forces the engine open. [Aircraft Commerce Issue 34, 2004; AJW and StandardAero mini-packs; Aircraft Value News, undated]

The life limits are the first of two clocks. The second is a thermometer.

EXHAUST-GAS TEMPERATURE (EGT) MARGIN

Every engine has a certified exhaust-gas temperature (EGT) red line it may not exceed. EGT margin is the gap between that limit and the temperature the engine actually reaches at full take-off thrust on a hot day. As blade tips and seals wear, the engine burns more fuel to make the same thrust and runs hotter, so the margin shrinks; it erodes fastest in the first 1,000 to 2,000 cycles after a shop visit. At zero margin the engine can no longer make its rated thrust within its limit and must come off the wing. Opening the core and restoring the high-pressure turbine's blades, vanes and seals recovers most of it. [Aircraft Commerce Issue 27, 2003; Part I §5]

TIME ON WING

Time on wing is the number of cycles or hours between an engine's installation and its removal. It is set by whichever of the two clocks runs out first: the life-limited part with the fewest cycles remaining, or the exhaust-gas-temperature margin, with a mandatory inspection as a rarer third. A mature CFM56 has a performance restoration every 10,000 to 15,000 cycles and a full overhaul with life-limited-part replacement at 20,000 to 25,000 cycles (Leeham, Mar 2024); at 1,000 to 1,500 cycles a year that is roughly seven to fifteen years between restorations. [Leeham News, 8 Mar 2024; Aircraft Value News, undated; Part I §5–§6]

WORKED EXAMPLE 5.1 — THE SHORTER CLOCK GOVERNS

Inputs: a CFM56-7B whose high-pressure turbine disk has consumed 18,500 cycles against its 20,000-cycle limit (limit sourced from the records mini-packs; cycles consumed assumed). The aircraft flies 1,500 cycles a year (sourced as the top of Aircraft Value News's range). The engine's temperature margin would allow another 5,000 cycles (assumed).

cycles remaining on the disk = $20,000 - 18,500 = 1,500$
time to the limit = $1,500 \div 1,500$ a year = about one year
margin remaining = 5,000 cycles, worth nothing to this owner

So the disk sets the removal date at about a year, and the engine comes off with 5,000 cycles of unused margin. The same arithmetic runs the other way on a fresh disk and a worn hot section: the margin governs and the disk's remaining life is carried into the shop unused. Limits of the example: two inputs are assumed; the point is the rule, not the figures. [Part I worked example 1.1]

Three consequences follow, and the rest of the chain is built on them. First, an engine at a shop visit is really three or four assets with three or four different remaining lives, because CFM set the limits at 30,000, 20,000 and 25,000 cycles and the margin clock runs on its own schedule. A fan and a core from the same engine are never at the same point in their lives. Second, the quickest way to return an engine to service is often to replace the one module that has run out rather than to restore all of them, because the engine was designed to split at the module flanges. Third, the usable supply of engines and modules is the documented supply, not the physical one.

A module with a gap in its records cannot be installed, so it does not count as supply however good its metal. The fleet behind all of this only shrinks: every engine retired, torn down or converted leaves the pool, and nothing enters it. [Part I §3–§4, §9]

Depth: Part I §1–§5.

S2. What a shop visit is and costs, and why the parts bill is where the money sits

When either clock runs out, or something breaks, the engine comes off and goes to a shop. The bill it generates there has five lines, and the mix among them decides most of what follows.

SHOP VISIT

A shop visit is any removal of the engine from the aircraft for work in an engine shop, from a visit to fix one finding to a complete rebuild. The shop disassembles the engine on stands, cleans and inspects every piece, replaces or repairs what is out of limits, rebuilds it and runs it in a test cell before release. "Teardown" is reserved in this primer for dismantling an engine to sell its parts, not for shop work. [Part I §6]

MAINTENANCE, REPAIR AND OVERHAUL (MRO), AND THE INDEPENDENT MRO

MRO is the industry's generic term for maintenance, repair and overhaul work and for the companies that do it. An independent MRO is an engine shop not owned by the engine's manufacturer: a stand-alone company such as StandardAero, an airline's maintenance arm that sells to others, or another engine maker's shop that is independent on this engine, such as MTU (a German engine maker) through MTU Maintenance. Every shop holds a regulatory repair-station approval and works to the manufacturer's shop manual. [Part I §6; Part IV §25]

WORKSCOPE, PERFORMANCE RESTORATION AND OVERHAUL

The workscope is the specification of what a visit will do: which modules are opened, how deep, and which parts are replaced new, repaired, or swapped for used material. A performance restoration opens the core and restores the compressor, combustor and high-pressure turbine to recover temperature margin; the fan and low-pressure turbine may be inspected but are not fully disassembled. An overhaul disassembles every module to piece parts and replaces the life-limited parts that would fall due within the next run; "zero-time" is one source's term for an overhaul with a complete new set. Replacing life-limited parts is a separate decision that a life-limit-driven removal combines with the restoration. A hospital or quick-turn visit fixes one thing and returns the engine; a module exchange is one of the things it does. [Part I §6; registry C-7]

TURNAROUND TIME (TAT) AND THE INDUCTION SLOT

Turnaround time (TAT) is the calendar time from an engine's induction into a shop to its release. A full CFM56 overhaul took about 60 days before the pandemic and 90 to 120 days in 2025 (trade press); GE's shops report "sustaining about 90 days" (May 2026); a consultancy blog gives 75 to 90 days for a restoration and 120 to 150 for a heavy visit. The induction slot is the date a shop commits to start an engine, and when shops are full the wait for a slot is additional to the turnaround: two to three months and up to six (Bain, 2024). Every day away needs a spare engine or grounds an aircraft. [Aviation Business News, 2025; GE Aerospace, 27 May 2026; Safe Fly Aviation, 2026 (consultancy blog); Bain, 2024; Part I §6; Part IV §27]

THE SHOP-VISIT COST BUILD

The cost of a shop visit is the sum of labour (hours times the shop's rate), new OEM parts that are scrapped and replaced (blades, vanes, seals, bearings), repairs of parts that can be restored rather than replaced, life-limited-part replacement (new or used), and the test-cell run, plus freight, consumables and the shop's margin. Material is 60 to 70% of the total, labour 20 to 30% and repairs 10 to 20% (Air Cargo Week); a consultancy blog adds that life-limited parts are 40 to 60% of parts cost when replaced and high-pressure turbine blades and vanes 15 to 25%. [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026 (consultancy blog); Part I §7]

CATALOGUE LIST PRICE AND ESCALATION

The catalogue list price is the manufacturer's published price for every new spare part, repriced at least annually. A shop passes it through at list plus a handling margin, and used and alternative parts are priced as discounts to it, so they inherit its movements. Escalation is the manufacturer's annual increase. Trade press put new spare-parts prices up around 10% a year in 2022 and 2023, with a double-digit step on 1 November 2022 and a high-single-digit step in August 2023; Safran's chief executive described a rule of pricing "above inflation by 3 to 4 points"; and the published prices of a full -7B life-limited-part set compounded at about 7% a year from 2008 to 2025, a figure derived in the research. The two series cover different parts and years and are carried together. [Aviation Week, late 2023; Part I worked example 1.3; Part IV §28; reconciliation R-018]

EXHIBIT S.2 — WHAT A CFM56 SHOP VISIT COSTS AND HOW LONG IT TAKES: EVERY BINDING RANGE WITH ITS SOURCE LEVEL

Quantity	Figure	Source and level	Note
Light or quick-turn visit	\$650–900k	Safe Fly Aviation, 2026 (consultancy blog)	No life-limited parts

Quantity	Figure	Source and level	Note
Performance restoration	\$1.2–1.6M on one page; \$1.8–2.5M on another	Safe Fly Aviation, 2026, two pages (consultancy blog)	Part IV uses the first range, Part I the second; never averaged (R-013)
Heavy visit with life-limited parts	\$2.1–2.8M on one page; ">\$3.5M" on another	Safe Fly Aviation, 2026 (consultancy blog)	Neither is consistent with a full new set at current list (see below)
Full overhaul ("zero-time")	\$3.5–4.2M	Safe Fly Aviation, 2026 (consultancy blog)	Full new set assumed by the label
Full -7B life-limited-part set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (June 2025, scope not visible)	Aircraft Commerce Issues 58 and 120; Aircraft Value News; MyAirTrade	Part IV computes with \$4M, Part I with \$5.7M (R-016)
Cost shares	Material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Air Cargo Week, 2025/26; Safe Fly Aviation, 2026	Two secondary sources; neither names shop data (R-014)
Turnaround	90–120 days full overhaul, ~45 light (trade press, 2025); ~90 sustaining (GE, May 2026); 75–90 restoration, 120–150 heavy (Safe Fly)	Aviation Business News; GE Aerospace; Safe Fly Aviation	Different shops and workscopes (R-019)
Slot wait before induction	Up two to three months, up to six	Bain, 2024 (consultancy)	Additional to turnaround
Spare-engine rent while away	~\$100,000/month per IBA (an appraiser), Apr 2024; \$42,000–48,000/month per Safe Fly Aviation (a consultancy blog), 2026, may be green-time	IBA; Safe Fly Aviation	Carried as a pair (R-028)

Source: Safe Fly Aviation, "Engine Shop Visit Costs Worldwide 2026", "What determines aircraft engine overhaul costs?" and "CFM56-7B engine availability report 2026" (consultancy blog); Aircraft Commerce Issues 58 (2008) and 120 (2018); Aircraft Value News, Nov 2019; MyAirTrade, June 2025; Air Cargo Week, "The true cost of engine maintenance", 2025/26; Aviation Business News, 2025; GE Aerospace, Bernstein conference, 27 May 2026; Bain, 2024; IBA via AviTrader, 25 Apr 2024. Reconciliation E.2 and rules 4 to 6. Accessed 2026-10-03.

Two things stand out in the exhibit, and the primer carries both without resolving them. Every cost figure comes from trade press or one consultancy blog; no manufacturer's or shop's price list

was read. And the two consultancy sets disagree with each other on the same scope. [reconciliation R-013] A further check comes from the set prices. A restoration at \$1.4M plus a full new set at \$4M is \$5.4M, above every quoted "with LLPs" range, so the quoted totals must assume partial replacement, used parts or older price levels. [Part I §7.1]

The life-limited-part line is where the arithmetic bites, because the parts are bought from one source at a price that only rises, and because an engine rarely reaches a shop with all of its parts at zero.

WORKED EXAMPLE S.2 — STUB LIFE, AND WHY THE LIFE-LIMITED-PART DECISION OUTWEIGHS THE WHOLE RESTORATION

Inputs: an engine arrives for a restoration at 12,000 cycles since new (assumed). Its core life-limited parts have a 20,000-cycle limit (sourced: mini-packs), so 8,000 cycles remain, and the owner plans a next run of 12,000 cycles (assumed, inside Leeham's 10,000–15,000 range). A full new set lists at \$5.7M (sourced: MyAirTrade, June 2025, scope not visible) or about \$4.0M (sourced: Aircraft Value News, Nov 2019); the two are never mixed in one line. The restoration itself costs \$2.15M (midpoint of Safe Fly's \$1.8–2.5M) or \$1.4M (midpoint of its \$1.2–1.6M).

keep the parts: the engine comes off again after 8,000 cycles, forced by the limit, with its restored performance two-thirds used
 replace them new now: 8,000 of each part's 20,000 cycles (40%) are scrapped unused
 cost per cycle, full life used: $5,700,000 \div 20,000 = \285 (at \$4.0M: \$200)
 cost per cycle, scrapped at 12,000: $5,700,000 \div 12,000 = \475 (at \$4.0M: \$333)
 restoration alone, per cycle of the next run: $2,150,000 \div 12,000 = \179 (at \$1.4M: \$117)

So the shorter remaining lives of the parts already inside the engine, the "stub life" (Aircraft Commerce), add a third to a half to the cost of every cycle if the parts are scrapped early, and that penalty is larger than the entire restoration workscope on either set price. The third option, which the whole used-parts market exists to supply, is to install used life-limited parts with 12,000 to 15,000 cycles remaining, matched to the planned run; dealers advertise such packages on cycles remaining, and their prices were not retrieved. Limits of the example: the cycles consumed and the planned run are assumed; the set price has two readings (R-016, R-017); the package price is unknown. [Aircraft Commerce Issue 34, 2004; Part I worked examples 1.2 and 1.6]

The parts bill is where the money sits for three reasons that the reader can now assemble. Material is 60 to 70% of a visit, and most material is priced off the manufacturer's catalogue, so the catalogue's annual escalation flows into every visit whoever performs it. Life-limited parts are 40 to 60% of parts cost when replaced, and no alternative maker produces one for the

CFM56, so that line is a single-source purchase at the escalating list price. And the high-pressure turbine's blades and vanes, the next largest line at 15 to 25% of parts cost, sit where most of the engine's lost margin is recovered, which is why an approved alternative to that blade matters as much as it does (S8). [Air Cargo Week, 2025/26; Safe Fly Aviation, 2026; Part I §7; Part VI §38] The levers a shop other than the manufacturer has are therefore all on the material line: used parts instead of new, approved alternative parts instead of the manufacturer's, repair instead of replacement, and exchange of a module instead of overhaul of the engine. Sections S8 and S9 take those four levers in turn.

Depth: Part I §6–§7; Part IV §29.

S3. Where the money sits across the chain: a first margin map, bases stated, nothing ranked

Before the chain is followed further, it helps to see its links and the profit each one reports, so that the segment sections have a frame. This is a map, not an analysis.

MARGIN POOL

The margin pool is this primer's term for how the profit earned in keeping an engine type flying is distributed across the links of the chain: the manufacturer's new parts and services, the shops' labour and test, the component repairers, the makers of approved alternative parts, the traders in used material and teardown, the lessors of engines and aircraft, and the servicers of third-party capital. No company reports a "margin pool"; the term names a way of reading the filings together. Its inputs are the slice-by-slice capture of a shop visit and each company's reported margin on its own basis. [Part I §11; Part XI §86]

FTAI'S TWO REPORTABLE SEGMENTS: AEROSPACE PRODUCTS AND AVIATION LEASING

FTAI Aviation reports two segments. Aerospace Products "develops and manufactures, repairs/refurbishes, and sells aircraft engines and aftermarket components primarily for the CFM56-7B, CFM56-5B, and V2500" through its Module Factory (S9); its FY2025 segment revenue was \$1,936.2M. Aviation Leasing "owns and manages aircraft and engines and leases and sells them, directly and through an equity-method investment"; it held 290 assets at 31 December 2025 (S10). A third line, FTAI Power, appears in the 2027 guidance; whether it is a reportable segment in the 2026 quarterly reports was not confirmed (S11). [FTAI 10-K FY2025; FTAI release, 29 Jul 2026; Part VII §45; Part VIII §58]

ADJUSTED EBITDA (CONSOLIDATED AND BY SEGMENT) AND GUIDANCE

Adjusted EBITDA is earnings before interest, taxes, depreciation and amortisation (EBITDA), as FTAI adjusts them: items the company treats as non-recurring or non-cash are removed, and the company's pro-rata share of Adjusted EBITDA from its unconsolidated equity-method investments is added back. FTAI reports it consolidated (\$597.3M in 2023, \$862.1M in 2024, \$1,190.9M in 2025) and by segment (Aerospace Products \$160.0M, \$380.6M, \$671.3M; Aviation Leasing \$500.1M and \$608.9M for 2024 and 2025), and it sets its guidance in the segment measure. The segment figures do not sum to the consolidated one; the difference is Corporate and Other, about −\$89.2M in 2025 by subtraction. The exact FY2025 definition and reconciliation were not retrieved. [FTAI 10-K FY2025 segment table; Part X §80, §84]

EXHIBIT S.3 — INDICATIVE MARGIN FIGURES ALONG THE CHAIN, EACH ON ITS OWN STATED BASIS

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
A shop visit, by slice	Any CFM56 visit	n/a	Material 60–70%; labour 20–30%; repairs 10–20% of the bill	Share of cost, not a margin	Air Cargo Week, 2025/26
Manufacturer: new engines, parts, services	GE Aerospace, Commercial Engines & Services	\$33,314M; services \$25,010M	26.6%	GE segment profit (GE's own measure)	GE 10-K FY2025
Approved alternative parts, with repair and distribution	HEICO (a maker of approved alternative parts), Flight Support Group	\$3,117.3M	24.1%	Segment operating income, derived; fiscal year to 31 Oct 2025	HEICO 10-K FY2025
Independent engine shop	StandardAero, Engine Services	\$5,354.0M	13.2%	Segment Adjusted EBITDA, StandardAero's definition	StandardAero 10-K FY2025
Independent engine shop (context, non-SEC)	MTU Aero Engines, commercial maintenance	€6.0B	8.0%	Adjusted EBIT (earnings before interest and tax, after depreciation)	MTU, 24 Feb 2026

Link in the chain	Company and segment	Revenue	Margin	Basis	Source
Independent engine shop (context, non-SEC)	Lufthansa Technik	€8.049B	7.5%	Adjusted EBIT	Lufthansa Technik, FY2025 results
Teardown and used material	AerSale, Asset Management Solutions	\$211.6M	35.0%	Gross margin, derived	AerSale 10-K FY2025
Module exchange, module and engine sales	FTAI, Aerospace Products	\$1,936.2M (FY2025); \$875.0M (Q2 2026)	34.7%; 28.5%	FTAI-defined segment Adjusted EBITDA, derived; includes related-party Maintenance, Repair and Exchange (MRE) Contract revenue (S9, S10)	FTAI 10-K FY2025; Q2 2026 release
Engine and aircraft lessor	FTAI, Aviation Leasing	Lease income \$235.2M (FY2025)	Segment Adjusted EBITDA \$608.9M; no margin computable on retrieved lines	Includes gains, maintenance revenue and the equity-method share	FTAI 10-K FY2025

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; GE 10-K FY2025; HEICO 10-K FY2025; StandardAero 10-K FY2025; MTU Aero Engines, FY2025 results, 24 Feb 2026; Lufthansa Technik, FY2025 results; AerSale 10-K FY2025; FTAI 10-K FY2025 and Q2 2026 release. Compiled from Part I Exhibit 1.14 and Part XI Exhibit 11.4. The five bases are not interchangeable and the fiscal years differ. Accessed 2026-10-03.

Three cautions travel with the table, and the reader who holds them can read every later figure correctly. The margins sit on five different bases. An EBITDA margin exceeds an EBIT margin by the depreciation share of revenue, which at FTAI was \$225.8M on \$2,507.4M in 2025, about nine points, so a lessor's EBITDA margin is high by construction and says little until depreciation and interest are deducted. [FTAI 10-K FY2025; Part X §80; Part XI §86] A share of cost is not a margin: the first row says where the bill goes, not what anyone earns on it. And a parts pass-through compresses a percentage: a shop that bills \$2M of the manufacturer's parts at a small handling margin beside \$1M of its own labour shows a thin percentage on a large revenue line, while a parts maker that sells its own product shows a wide one, and the two may earn similar dollars.

FTAI's segment sells a refurbished module or engine it owns, so its margin is closer in construction to a parts business's than to a shop's, and the research warns against reading it beside StandardAero's as the same activity measured twice. [Part XI §86.9]

One structural fact about the manufacturer's capture belongs on this map. Safran describes about 85% of the CFM56's aftermarket revenue as spare parts and time-and-materials work, against about 40% for the LEAP, and about 15% of the CFM56 fleet is under long-term flight-hour agreements against about 50% of the LEAP fleet. [Safran via Aviation Week, c. 2023/24; Aviation Week, undated; Part IV §28] On this engine the manufacturer earns mainly by selling parts at list into visits it does not perform and does not control. That open structure is what leaves room for every substitute described in S8 and S9; on a contracted engine the manufacturer chooses the parts, and it has no reason to choose a competitor's.

Depth: Part I §11; Part XI §86.

S4. Who owns engines and how they pay: leases, reserves, engine values, green time

Almost every dollar an engine earns passes through a contract, usually an operating lease, and almost every dollar its owner borrows is secured on that contract. The lease is where the cost of the life an engine consumes is turned into cash, and where an engine's value is written down part by part.

OPERATING LEASE

An operating lease is a contract under which the owner, the lessor, rents an aircraft or engine to an operator, the lessee, for a fixed term in exchange for periodic rent, while the lessor keeps title and keeps the risk that the asset is worth more or less than expected when it comes back. Nearly all of FTAI's leases are operating leases. At the end of the term the lessor extends, moves the asset to a new lessee, sells it, or parts it out. [FTAI 10-K FY2025; Part II §12]

MAINTENANCE RESERVES

Maintenance reserves are payments the lessee makes to the lessor in addition to rent, set per flight hour or per cycle and collected per component: each engine's performance restoration, each engine's life-limited parts, the airframe's heavy checks, the landing gear. They pre-fund the eventual shop visit, so that the lessor holds cash for the life consumed if the lessee defaults or hands the asset back. The lessor reimburses a completed event up to the lesser of the balance held and the qualifying cost; an unclaimed balance stays with the lessor at lease end. Strong-credit lessees may instead pay end-of-lease compensation in one lump for the difference in condition between delivery and return. On a mid-life narrowbody the reserve stream can approach half of basic rent, and the lessor books reserves as revenue when it is no longer obliged to return them, which is FTAI's "maintenance revenue" line (\$218.5M in 2025). [AerCap 20-F FY2013; FTAI 10-K FY2025; Part II §13]

RETURN CONDITIONS

Return conditions are the contractual minimum state of a leased engine at redelivery: minimum hours and cycles to the next restoration, minimum cycles remaining on each life-limited part, a borescope inspection and a test-cell run, and records in order. Together with the reserve or compensation regime they decide who pays for the maintenance life consumed during the lease. They may also exclude parts and repairs that are not the manufacturer's, which is where a lessor's view of alternative parts is written down; current practice on that clause for 2024 to 2026 was not found. [Part II §13; Part VI §42]

LEASE RATE FACTOR (LRF)

The lease rate factor (LRF) is the monthly rent divided by the asset's current market value, as a percentage; 1.0% a month is a 12% gross annual yield before depreciation, interest and maintenance. No source the research reached states a factor for FTAI, for Willis Lease or for the CFM56 as a type. Every figure in the primer is derived and carries its basis: 0.70% a month (a consultancy green-time rent on an appraiser's half-life value, mixed bases), 0.87 to 1.1% (FTAI's own 2025 book, on cost and on net book value), about 1.0% (Willis Lease's on-lease book, derived from its filing), and 0.9% assumed in one fee example. [Part II §12, §14; reconciliation R-030]

HALF-LIFE VALUE AND MAINTENANCE-ADJUSTED VALUE

Half-life is an appraisal convention, not a physical midpoint: the value of an engine with every module halfway between restorations and every life-limited part at half its certified life, so that no adjustment has been made for actual maintenance status (ISTAT). Appraisers such as IBA publish values on that basis. Maintenance-adjusted value is the half-life value plus, for each item, the actual remaining fraction less one half, multiplied by the cost to restore that item, summed module by module. On this primer's inputs the swing between a fresh and a run-out -7B of the same age is about \$2.8M on one set price and about \$1.9M on the other. Because the modules carry clocks that deliberately do not align, the adjustment is naturally computed module by module. [ISTAT Jetrader, 2010; IBA, Sept 2025; Part I §8]

GREEN TIME

Green time is the remaining usable life on an engine or module before its next mandatory shop visit, set by whichever runs out first: the life-limited part with the fewest cycles remaining, the exhaust-gas-temperature margin, or a hard inspection. A green-time engine is one the owner does not intend to restore, flown or leased as-is to use up that life and then torn down; a green-time lease rents that life for months to a few years at a rent that is the price of the time consumed. The pool of green time is fed by retirements, so it shrinks exactly when a slow-retiring fleet most wants it. [Part II §16; registry C-2]

EXHIBIT S.4 — CFM56-7B VALUES AND RENTS: EACH FIGURE WITH ITS BASIS, SOURCE LEVEL AND DATE

Quantity	Figure	Basis	Source and level	Date
-7B24 and -7B27 market value	\$5.7M; \$6.4M	Half-life, H2 2025	IBA (an appraiser), Engine Values 2025B	Sept 2025
Value direction	"Levelled off after approximately 18 months of increases"; rose about 20% from 2023 to 2024	Market value	IBA, H1 2026 update; IBA, Apr 2024	2026; 2024
-7B "green-time value"	\$2.8–3.4M	Labelled "(new condition)", which is internally inconsistent	Safe Fly Aviation (consultancy blog)	2026
Run-out -7B as teardown feedstock	Bought \$0.8–1.2M; yields \$1.6–2.2M of used material; net \$0.4–0.8M	Exhausted engine	Safe Fly Aviation (consultancy blog)	2026

Quantity	Figure	Basis	Source and level	Date
-7B spare-engine rent	~\$100,000 a month, from ~\$75,000 in 2019	Spare engine; condition not stated	IBA, via AviTrader	Apr 2024
-7B rent, alternative reading	\$42,000–48,000 a month; "about 34% above 2019"; up 15–20% "in recent months"	Beside green-time values; may be green-time; basis not stated	Safe Fly Aviation (consultancy blog)	2026
12-year-old 737-800 rent	~\$228,000 a month, down 11% year on year	Whole aircraft	IBA, via India Seatrade News	Jul 2025
10-year-old 737-800 rent	+44% against January 2022	Whole aircraft	Cirium Ascend via Bloomberg	2023–24 context

Source: IBA, "Engine Values Release September 2025 (2025B)"; IBA, "Engine and Lease Rate Update H1 2026"; IBA via AviTrader, 25 Apr 2024; Safe Fly Aviation, "CFM56 Engine Market Report 2026" (consultancy blog); India Seatrade News, 6 Sep 2026 (IBA's July 2025 commentary); Bloomberg via AJOT (a trade publication). Compiled from Part I Exhibit 1.11 and Part II Exhibit 2.3. Reconciliation R-028, R-029, R-031; rule 6. Accessed 2026-10-03.

The exhibit holds two disagreements that reach every later Part, and the primer carries both sides. On rent, an appraiser quoted about \$100,000 a month for a -7B spare engine in April 2024, and a consultancy blog quoted \$42,000 to \$48,000 in 2026 on a page that also lists green-time values, so the lower rent may price a part-worn engine although the source does not say so; the two agree on the change since 2019, about a third, and disagree on the level. [IBA, Apr 2024; Safe Fly Aviation, 2026; reconciliation R-028] On value, the appraiser gives half-life market values of \$5.7M and \$6.4M and says values have levelled off, while the blog gives a "green-time value" of \$2.8 to \$3.4M and rents still rising; the figures sit on different bases and disagree on direction. [IBA, Sept 2025 and H1 2026; Safe Fly Aviation, 2026; reconciliation R-029] The direction of whole-aircraft rents is contested too: a new-aircraft lessor's filing reports rates rising in 2025, and the appraiser reports a 12-year-old 737-800 down 11% in July 2025 and warns of a fall in September 2026. [Air Lease 10-K FY2025; IBA via India Seatrade News, 6 Sep 2026; reconciliation R-031]

WORKED EXAMPLE S.4 — TWO ENGINES AGAINST ONE AIRCRAFT: WHY THE ENGINES CARRY A MID-LIFE NARROWBODY'S VALUE

Inputs: a CFM56-7B spare engine rents for about \$100,000 a month (sourced: IBA, Apr 2024) or \$42,000–48,000 (sourced: Safe Fly Aviation, 2026, may be green-time); a 12-year-old 737-800 rents for about \$228,000 a month (sourced: IBA, Jul 2025). A 737-800 carries two engines.

on the appraiser's engine rent: $2 \times 100,000 = \$200,000$ a month = 88% of \$228,000; \$28,000 (12%) left for the airframe and everything else
on the consultancy's engine rent: $2 \times 42,000$ to $2 \times 48,000 = \$84,000$ –
96,000 = 37% to 42% of \$228,000

On the appraiser's figure a pair of engines on standalone leases would command 88% of what the whole aircraft commands; on the consultancy's figure, 37 to 42%. The first mixes a 2024 engine rent with a 2025 aircraft rent, and the second may price part-worn engines; the primer carries both. What neither disputes is the direction. A pair of engines earns a large share of a mid-life narrowbody's rent, and a pair of fresh ones earns most of it. Four facts from S1 explain why: the engines are separable; their value is dominated by maintenance condition; no new CFM56 is being built while the airframes are being replaced; and the airframe at 15 to 20 years has few other uses, so a 21-year-old 737-800 rents for "less than \$160,000" while a -7B rents for the same whether it hangs on a 12-year-old wing or a 21-year-old one. FTAI states the consequence in its own words: it "owns and leases jet aircraft which often facilitates the acquisition of engines at attractive prices". Limits: mixed dates and bases; the two engine rents are a contested pair; no appraised value for a 12-to-15-year-old 737-800 was retrieved. [IBA, Apr 2024 and Jul 2025; Safe Fly Aviation, 2026; FTAI Q2 2026 release; Part II worked example 2.6]

The floor under an engine's value is what it fetches taken apart, which S8 works. Between a fresh engine at its half-life value plus an adjustment and a run-out engine worth its parts lies the market in time: a part-worn engine leased for its remaining months, or torn down now. That choice, and the pool of engines it draws on, is where a slow-retiring fleet shows up first.

SPECIAL-PURPOSE VEHICLE (SPV), ASSET-BACKED SECURITIZATION (ABS) AND WAREHOUSE FACILITY

A special-purpose vehicle (SPV) is a bankruptcy-remote entity that exists only to own a defined pool of assets and the debt secured on them, so a lender looks only to those assets and their leases, not to the sponsor. An asset-backed securitization (ABS) is a sale of a portfolio of leased aircraft or engines to such a vehicle, which issues notes to investors in tranches paid in order of seniority; the sponsor usually keeps the equity and services the leases for a fee, and rating agencies size the senior notes by stressing rents, downtime, maintenance and residual values. A warehouse facility is a revolving bank loan to an SPV that funds assets one at a time against an advance rate until there are enough to securitize. FTAI's capital vehicles, under its Strategic Capital Initiative (SCI), are built from all three (S10); on 22 May 2026 the first of them priced FTAI MRE 2026-1, \$612M of notes on 48 A320ceo and 737NG aircraft on lease to 23 airlines, with Series A expected to be rated Asf by Fitch and A(sf) by KBRA (a rating agency) and Series B BBB+sf by Fitch. [FTAI release, 22 May 2026, via Finviz and AviTrader; Part II §17]

The reason the rating agencies matter beyond the lessor's own income statement is now visible. An appraiser's note that engine values have levelled off, or that narrowbody rents fell 11 to 13%, feeds straight into the stress tests that decide how much senior debt a pool of leased engines can carry. [Part II §17]

Depth: Part II §12–§17.

S5. Why 2026 looks the way it does: the fleet that only shrinks, retirements that did not come, new engines that did not relieve, airframers that did not deliver

Nothing in S1 to S4 changes between a quiet year and a tight one. What changes is the quantities: how many engines need a shop visit, how many are torn down, how many spares are needed and for how long. Between 2019 and 2026 four things moved those quantities in the same direction.

INSTALLED BASE, RETIREMENT RATE AND THE SHOP-VISIT FORECAST

The installed base is the population of engines fitted to aircraft in the operating fleet; published CFM56 counts of about 23,000 (CFM, undated), about 24,000 (Aviation Business News, Sept 2025) and more than 22,800 (Safran, end-2025) may include spare and stored engines, and a consultancy blog's about 14,200 "active" and more than 19,000 "flying" are labelled consultancy figures and never the fleet figure. The retirement rate is the share of a fleet withdrawn in a year; GE's planning assumption for CFM56 retirements in 2026 moved from 3–4% of the base, to 2–3%, to 1.5–2% (GE's chief financial officer via Aviation Week, 2026), GE reported about 1.5% retired in 2025 and expected about 2% in 2026 (via Leeham, 22 Jan 2026), and first-quarter 2026 retirements ran "below 1%" (GE investor-relations page). The shop-visit forecast counts expected inductions a year: 2,300 to 2,400 CFM56 visits a year in 2026 to 2028 with the apex around 2027 to 2028 and a gradual fade (GE via Aviation Week, 2026), against an earlier Safran framing of a peak of about 2,500 in 2025 to 2026 (c. 2023/24), from just above 2,000 in 2019. [Part III §18, §19, §23; reconciliation R-001, R-004, R-006]

Because no new -5B or -7B is built, the base falls by exactly the number of engines retired or converted each year. Every engine that does not retire keeps flying, keeps accumulating cycles, and reaches the visits of S2. Every engine that does not retire is also one that is not torn down, so its parts do not reach the used-material market. The same number sits on both sides of the aftermarket at once.

EXHIBIT S.5 — THE CONTEXT IN ITS BINDING FIGURES, EACH WITH SOURCE, LEVEL AND DATE

Driver	Figure	Source and level	Date
CFM56 in service	~23,000; ~24,000; >22,800 (OEM-level); ~14,200 and >19,000 (consultancy, labelled)	CFM website; Aviation Business News; Safran; Safe Fly Aviation	Undated; Sept 2025; end-2025; 2026
CFM56 retirement rate	Planning 3–4% → 2–3% → 1.5–2%; 2025 actual ~1.5%; 2026 expected ~2%; 1Q 2026 "below 1%"	GE CFO via Aviation Week; GE via Leeham; GE investor-relations page	2026; 22 Jan 2026; 2026

Driver	Figure	Source and level	Date
Aircraft retirements	502 / 465 / 333 / 325 in 2019–22 (Cirium); 2024: "fewer than 200" (IBA) and 587 (Cirium via Informa); scope not visible in either	Cirium via Aerodynamic Advisory; IBA via AviTrader; Informa	2023; Sept 2024; undated
CFM56 shop visits a year	Just above 2,000 (2019); 2,300–2,400 through 2028, apex 2027–28 (GE); ~2,500 peak 2025–26 (Safran, earlier)	Safran and GE via Aviation Week	c. 2023/24; 2026
Pratt & Whitney geared-turbofan groundings	835 PW1100G-powered A320neos grounded or stored (late 2025); RTX (Pratt & Whitney's parent) planned ~350 on average over 2024–26; aircraft-on-ground (AOG) levels down more than 20% from the 2025 peak	Cirium via The Flying Engineer; RTX via Leeham	Late 2025; 27 Jan 2026
LEAP durability	Early removals at 2,000–6,000 cycles against a mature CFM56's 10,000–15,000; LEAP-1B durability kit certified 18 Jul 2026, production cutover early 2027	Visual Approach; Leeham; CFM release	Undated; 2024; 18 Jul 2026
737 MAX production cap	38 a month from Jan 2024; 42 from Oct 2025; review to 47 completed May 2026; 2018 peak 52; deliveries 260 (2024), 440 (2025)	Federal Aviation Administration (FAA) via TravelPulse; CAPA (the Centre for Aviation); Wikipedia compilation	2024–26
Airbus deliveries	793 in 2025 against 820 originally targeted; A320neo rate 75 only by end-2027	Argus Media; Air Data News	2025–26
Lessor extensions	75–90% of scheduled lease expiries extended; about 12 mid-life 737NGs and fewer than 25 A320ceos available to rent worldwide; extension terms 4–6 years	Cirium Ascend via Bloomberg; IBA	2023–24 context; Nov 2023

Source: as listed in each row; CFM International; Aviation Business News, Sept 2025; Safran, 13 Feb 2026; Safe Fly Aviation, 2026; Aviation Week, "CFM56 Overhaul Demand Remains Strong", 2026, and "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24; Leeham News, 22 Jan 2026 and 27 Jan 2026; GE Aerospace investor-relations page, 2026; IATA (the airlines' trade body) Maintenance Cost Conference, 2023; AviTrader, 24 Sep 2024; Informa Markets forecast page; The Flying Engineer, 2026; Visual Approach; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA, May 2026; Argus Media; Air Data News; Bloomberg via AJOT; IBA via Key.Aero. Compiled from Part III Exhibits 3.1, 3.5 to 3.8, 3.10 to 3.12. Accessed 2026-10-03.

The fleet count is itself contested, and the primer carries the disagreement because every per-engine figure in the chain divides by it. The three manufacturer-level figures sit close together and none states its basis. The consultancy blog's 14,200 was read by one dossier as the whole active fleet and by another as the -7B fleet alone, and its own ">19,000 flying" cannot be reconciled with it from the material reached. [reconciliation R-001, R-002] The retirement record is contested twice over. GE's own planning number for 2026 fell three times within a year, and the realised rate fell further; the two venues that report it differ by half a point on the 2026 expectation. At aircraft level, an appraiser said fewer than 200 aircraft retired in 2024 and a data provider recorded 587, a factor of three, with the scope visible in neither; arithmetic on other sources gives about 410 a year and about 1,130 a year for roughly the same quantity. The primer presents all four and stops. [GE CFO via Aviation Week, 2026; Leeham, 22 Jan 2026; IBA via AviTrader, 24 Sep 2024; Cirium via Informa; reconciliation R-004, R-005]

WORKED EXAMPLE S.5 — WHAT THE GAP BETWEEN THE PLANNED AND THE REALISED RETIREMENT RATE MEANS IN ENGINES

Inputs: installed base about 24,000 (sourced: Aviation Business News, Sept 2025) or more than 22,800 (sourced: Safran, end-2025); rates taken as the midpoints of GE's stated ranges, 3.5% for "3–4%" and 1.75% for "1.5–2%" (midpoints are the primer's choice; ranges sourced: GE CFO via Aviation Week, 2026); the 2025 actual of about 1.5% and the 2026 expectation of about 2% (sourced: GE via Leeham, 22 Jan 2026).

on 24,000: 3.5% = 840 engines a year; 1.75% = 420; 1.5% = 360; 2.0% = 480; below 1% = fewer than 240
 on 22,800: 3.5% = 798; 1.75% = 399; 1.5% = 342; 2.0% = 456
 gap between the first plan and the realised range: $24,000 \times (0.035 - 0.0175) = 420$ a year; over 2024–26, about 1,200 to 1,260

So about 400 engines a year on any manufacturer-level base, roughly 1,200 over three years, are still flying that the manufacturer's first plan had retired. Each is a future shop visit and a life-limited-part set that will reach its limit, and each is a teardown that did not happen. Against that, the manufacturer's shop-visit range widened by about 100 visits, from "about 2,300" to "2,300–2,400", because a retained engine visits a shop only when its own clocks run out. Limits: the base is undefined as to spares and stored engines; the midpoints are the primer's; the three-year figure is an upper bound because the base shrinks a little each year. [Part III worked example 3.2]

Why did so few engines retire? An aircraft is retired when its next heavy event, a shop visit or a life-limited-part replacement, costs more than it is worth kept, and when a replacement is available. Between 2023 and 2026 the second condition failed. An airline at the maintenance cliff with no new aircraft to fly pays for the shop visit anyway, because the alternative is to shrink; a lessor whose lessee cannot find a replacement extends the lease rather than take the aircraft back into a market with no buyer-operators. Cirium's count that only about a dozen mid-life 737NGs

and fewer than 25 A320neos were available to rent anywhere in the world is the measure of how closed the gate was. [Cirium Ascend via Bloomberg; Part III §19] Two things closed it.

THE CONTEXT SHOCK: THE GEARED TURBOFAN (GTF) POWDER-METAL ISSUE, LEAP DURABILITY AND ITS KIT, AND THE 737 MAX RATE CAP

Three failures of replacement kept the old fleet flying. Pratt & Whitney's geared turbofan (GTF), one of two engines on the A320neo, had turbine discs made from contaminated metal powder; RTX (Pratt & Whitney's parent) disclosed the defect in 2023, and 835 PW1100G-powered A320neos were grounded or stored in late 2025 on Cirium's count against RTX's planning figure of about 350 on average, with aircraft-on-ground levels down more than 20% from the 2025 peak by January 2026 (RTX via Leeham). CFM's LEAP, on the A320neo and sole on the 737 MAX, came off wing early at 2,000 to 6,000 cycles in hot and harsh environments; CFM certified a LEAP-1B durability kit with the Federal Aviation Administration (FAA) and the European Union Aviation Safety Agency (EASA) on 18 July 2026, with full production cutover expected in early 2027. And the FAA capped 737 production at 38 a month after the January 2024 door-plug event, raising it to 42 in October 2025 and completing the review to 47 in May 2026, against a 2018 peak of 52; Airbus delivered 793 aircraft in 2025 against an original target of 820 and now expects A320neo rate 75 only by the end of 2027. [Cirium via The Flying Engineer, 2026; Leeham, 27 Jan 2026; CFM release, 18 Jul 2026; TravelPulse, Oct 2025; CAPA, May 2026; Argus Media; Air Data News; Part III §20–§21]

The mechanism from each shock to the CFM56 fleet is indirect and specific. A grounded Pratt & Whitney-powered A320neo needs its flying replaced, and the practical replacement is an A320ceo on the same pilot rating; that keeps the -5B and the V2500 flying harder, not the -7B. An early LEAP removal generates a LEAP shop visit in the manufacturer's network, not a CFM56 visit, but an operator whose 737 MAX fleet is less available than planned keeps its 737NGs; that touches the -7B. A production cap of 38 a month against a 2018 rate of 52 holds back on the order of 170 737NG replacements a year at a one-for-one ceiling, or about 77 at Cirium's replacement ratio of 46.1%. [Part III §20.3, worked example 3.4] Each shock also carries its own reversal: the GTF groundings were already easing in January 2026, the LEAP-1B kit cuts over in early 2027, and the backlogs are owed rather than cancelled, so when the rates rise the deferred retirements arrive with them.

The lessors' response closed the loop. With 75 to 90% of scheduled expiries extended for four to six years, the maintenance events that would have fallen at a lease transition fell inside the extended term and were paid by the incumbent airline, and only 10 to 25 aircraft of every 100 came back to the used market as acquisition candidates or teardown feedstock. [Cirium Ascend via Bloomberg; IBA, Nov 2023; Part III worked example 3.5] The shape of the forecast that results is contested between the two CFM partners at two dates, a peak of about 2,500 in 2025 to 2026 against a plateau of 2,300 to 2,400 through 2028, and the long end of either curve depends on the retirement rate that moved three times in a year. [reconciliation R-006, R-007]

Depth: Part III §18–§23.

S6. The translation table in one page: each driver, the requirement it creates, what it does not stress

Part III turns each documented driver into a numbered row stating the requirement it creates in the aftermarket's own units and what it leaves untouched or pushes the other way. The rows are the organising device of the primer: every segment Part opens by naming the rows it answers. The last column matters as much as the middle one. A driver that raises shop-visit demand but removes teardown feedstock has done two things, and a reader who carries only the first will misread the used-parts market.

EXHIBIT S.6 — THE TRANSLATION TABLE: FOURTEEN DRIVERS, THE REQUIREMENT EACH CREATES, AND WHAT EACH DOES NOT STRESS

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-1	Low CFM56 retirements: ~1.5% in 2025, ~2% expected for 2026, "below 1%" in 1Q 2026, against a 3–4% plan (GE)	More engines accruing cycles: sustains 2,300–2,400 visits a year through 2028; more LLP replacements and full restorations in place of green-time run-outs; module demand	Starves teardown feedstock (T-9); does not change shop capacity; no effect on LEAP or GTF spares
T-2	GTF powder-metal groundings: 835 aircraft grounded or stored (Cirium, late 2025); easing from the 2025 peak (RTX)	A320ceo lift to replace grounded neos: ceo extensions and short-term leases; V2500 and -5B visits brought forward; spare and green-time -5B and V2500 modules	No -7B demand except through general scarcity; adds no used-material supply; creates no CFM56 work on LEAP-type engines
T-3	LEAP durability: removals at 2,000–6,000 cycles; LEAP-1B kit certified 18 Jul 2026, cutover early 2027	LEAP visits in the manufacturer's network under long-term agreements; indirectly, slower ceo and 737NG retirements	Creates no CFM56 visits, modules or used material; the indirect support fades after the kit cuts over

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-4	737 MAX production cap: 38 → 42 → 47 a month; 260 and 440 MAX delivered in 2024 and 2025	737NG extensions and high utilisation: -7B visits, LLP replacements, spare engines and green-time modules	Does not touch -5B or V2500 demand; reverses as the cap lifts toward 52, with more 737NG retirements and -7B teardown supply later; United, WestJet and Copa retirements are the counter-case
T-5	Airbus shortfall: 793 against 820 in 2025; rate 75 only by end-2027	A320ceo extensions and utilisation: -5B and V2500 visits and spares; keeps ceo part-out low	Does not stress the -7B; converts planned capital spending on new aircraft into aftermarket spending on old ones; reverses as the rate rises
T-6	Lessor extensions: 75–90% of expiries; ~12 737NGs and <25 A320ceos available; terms 4–6 years	Maintenance paid by the incumbent inside the extended term; higher mid-life lease rates; capital held in aircraft on lease	Only 10–25 of every 100 aircraft return: fewer acquisition candidates and less feedstock; does not change the cycles the fleet flies; rent direction contested (R-031)
T-7	OEM parts escalation: ~10% a year in 2022–23; inflation plus 3–4 points (Safran); the -7B LLP set ~7% a year compounded 2008–25	Demand for substitutes on the material line, 60–70% of every visit: used material, approved alternative (PMA) parts, repair, module exchange; raises the dollar value of every visit	Does not change the removal count; does not stress labour or slots; leaves the LLP line unsubstituted, since no PMA exists for a CFM56 LLP
T-8	Capacity shortfall: slot waits 2–6 months; overhaul 90–120 days; demand to exceed capacity by >17% by decade end (Bain); two-thirds of operators short of technicians (Oliver Wyman)	Spare engines and leases in proportion to the delay; module exchanges and quick-turns as substitutes for full overhauls; inventory held for exchange	Does not change removals; adds no teardown supply; GE reports turnaround improving (R-019, R-020)

Row	Driver and its binding figure	Requirement it creates	What it does not stress, or pushes the other way
T-9	Used-material scarcity: supply "remains thin" (Aviation Maintenance magazine, 2025); AerSale whole-asset sales 20 engines → 13; values up as much as 50% in two years (ePlaneAI)	Higher used-part and green-time prices; more new parts bought; PMA as the other substitute; capital to acquire feedstock, including whole aircraft bought for engines	A consequence of T-1, T-4, T-5 and T-6, and reverses when they do; adds no capacity
T-10	Data-center power and the turbine shortage: data-center electricity to ~945 TWh by 2030 (International Energy Agency (IEA), Apr 2025); GE Vernova backlog plus reservations 116 gigawatts (GW) at Q2 2026, slots sold for 2031; Siemens Energy lead times three-plus years	A new sink for CFM56 cores outside aviation: 100 Mod-1 units in 2027 would consume about 100 engines a year on one core per unit	Creates no aviation visits and no used material; 100 engines is 21–28% of a year's retirements on an OEM-level base at 1.5–2%, about 35% on the consultancy base; engines per unit undisclosed
T-11	Freighter conversion: >260 737-800 freighters in service, >300 in the pipeline; IBA warns of "oversupply of narrowbody conversions" (6 Sep 2026)	Green-time -7B modules and visits for 20-plus-year-old airframes	Removes airframes and engines from the teardown pool; cargo-market dependent and contested (R-034)
T-12	Higher utilisation of mature narrowbodies, "rising significantly" (IBA, Sept 2024)	Visits and LLP replacements earlier in calendar time; faster margin erosion	Creates no used material; ends when replacements arrive
T-13	GTF groundings easing: AOG down 25% year to date, output up 40% (RTX, Jan 2026)	Returning neos reduce ceo lift: ceo extension and lease rates soften; more -5B and V2500 teardown supply later	Does not reduce the ceo visits already induced; full resolution put at 2027–28
T-14	CFM56 plateau then decline: 2,300–2,400 through 2028, apex 2027–28, gradual fade (GE); ~2,000 CFM56 visits by 2030 implied (Safran); LEAP aftermarket revenue parity in 2028	Long run: shop capacity moves to the LEAP; teardown supply rises as retirements normalise (~2.7% of the aircraft fleet a year from 2028, McKinsey)	Does not stress 2026–27 demand; timing depends on a retirement rate revised three times in one year

Source: Part III Exhibit 3.14 and the sources cited there for each row; rows T-7, T-9 and T-10 draw on Part I §4, Part V §33 and Part IX §71. Reconciliation R-001, R-004, R-006, R-007, R-019, R-020, R-028, R-031, R-034. Accessed 2026-10-03.

Read down the right-hand column and the chain's central tension is visible before any segment is described. The drivers that raise the demand for shop visits, modules and spare engines (T-1, T-4, T-5, T-6) are the same drivers that cut the supply of used parts and green-time engines (T-9), because the engine that is not retired is both. The one driver that acts on price rather than quantity (T-7) raises the dollar value of every visit and creates the demand for everything that substitutes for a new part. And the one driver that removes engines from aviation for a reason unrelated to aviation (T-10) competes with the teardown trade and the exchange business for the same run-out cores. The rows are answered as follows: T-1, T-7 and T-8 by the shop-visit market (S7); T-1, T-6, T-7, T-9 and T-11 by used material and alternative parts (S8); T-1, T-2, T-4, T-7 and T-8 by the Module Factory (S9); T-2, T-4, T-5, T-6, T-8 and T-9 by leasing and capital (S10); T-10 by FTAI Power (S11). [Part III §24, Exhibit 3.15]

Depth: Part III §24.

S7. The shop-visit market and the OEM's pricing: demand plateau, capacity shortage, list-price escalation, the OEM's own offerings

The market that S1's engine creates has four kinds of shop and one price-setter. The shops are the manufacturer's own network, the large independents, the airlines' maintenance arms and, since 2022, FTAI's Module Factory. The price-setter is the manufacturer, which earns from nearly every visit without performing it.

The demand side is a plateau whose height and length are contested. Safran's earlier framing, from about 2023 or 2024, had CFM56 visits peaking at about 2,500 a year in 2025 and 2026 and then falling. GE's 2026 framing has about 2,300 for each of 2026 and 2027, widened to 2,300 to 2,400 and "probably closer to 2,400", held through 2028 with the apex around 2027 to 2028 and a gradual fade after. The later view is both lower at its top and later in reaching it. [Safran and GE via Aviation Week; reconciliation R-006, R-007] The CFM56 is roughly a quarter of all engine shop work by count: about 2,300 visits inside about 9,000 of all types in 2025. [Aviation Week 2026 Fleet & MRO Forecast, via ePlaneAI] What moved the shape between the two dates was not the fleet, which only shrinks, but the retirement assumption behind it, which is why a shop or a module factory adding capacity now is adding it into a plateau on the manufacturer's own forecast, and why the date the fade begins is the least certain number in the chain.

WORKED EXAMPLE S.7 — RECONCILING 2,300 VISITS A YEAR WITH THE INSTALLED BASE, AND WHY THE BASE IS THE CHAIN'S LARGEST UNKNOWN

Inputs: 2,300 visits a year (sourced: GE via Aviation Week, 2026, low end); bases of about 24,000 (trade press), more than 22,800 (Safran), more than 19,000 "flying" and about 14,200 "active" (consultancy blog, labelled); a mature restoration every 10,000–15,000 cycles at 1,000–

1,500 cycles a year (sourced: Leeham, Mar 2024; Aircraft Value News), roughly 6.7 to 15 years; a spare ratio of about 10% of installed engines (sourced: MTU).

on 24,000: $2,300 \div 24,000 = 9.6\%$ a year $\rightarrow$ about 10.4 years between visits
 on 22,800: $10.1\% \rightarrow 9.9$ years; on 19,000: $12.1\% \rightarrow 8.3$ years; on 14,200:
 $16.2\% \rightarrow 6.2$ years
 removing 10% spares from 24,000: $2,300 \div 21,600 = 10.6\% \rightarrow 9.4$ years

An implied interval of about ten years matches the mature interval only at the long end of its range; about six years on the consultancy base is shorter than any sourced interval unless engines fly more than about 1,600 cycles a year, above the sourced range. So the 24,000 base is consistent with the sourced interval only if it includes a few thousand spare, stored or green-time engines, or if the long first runs of a still-maturing fleet are depressing the visit rate; about 70% of the fleet had seen at most one shop visit in 2024 (Safran). Which base is right decides every per-engine figure in the chain, including FTAI's share of it. Limits: the visit count and the base come from different publishers and may define "CFM56" differently; the spare ratio is an industry generality. [Part III worked example 3.6; reconciliation R-001, R-008]

The capacity side has two readings and the primer carries both. Trade press and consultancies describe a system that is full: a CFM56 overhaul at 90 to 120 days against about 60 before the pandemic, waits for an induction slot of two to six months, engine-maintenance demand set to exceed capacity by more than 17% by the end of the decade if capacity grows on its current path (Bain, 2024), two-thirds of maintenance operators unable to find enough technicians and labour-rate inflation of 5.5 to 6.0% (Oliver Wyman, April 2026), and used-material supply that "remains thin". GE's own filing and conference material read the other way: turnaround for the LEAP, CFM56 and GE90 (a widebody engine) improved by more than 10% year on year in the fourth quarter of 2025, material input from priority suppliers rose more than 40%, and CFM56 turnaround was "sustaining about 90 days" in May 2026. [Aviation Business News, 2025; Bain, 2024; Oliver Wyman, Apr 2026; GE 10-K FY2025; GE Aerospace, 27 May 2026; reconciliation R-019, R-020] The consequence for the chain does not depend on which reading is right. A removal that returns in five to ten months ties up a spare for that long, so the shortage of slots is itself demand for spare engines and spare-engine leases, and for anything that returns the aircraft to service in weeks rather than months, which is the exchange of S9.

The manufacturer's model is the frame everything else sits in.

EXHIBIT S.7 — THE OEM AFTERMARKET MODEL ON THE CFM56: HOW THE MANUFACTURER EARNs FROM A VISIT IT DOES NOT PERFORM

Element	What it is	Binding figure	Source and level
Spare parts at list	New parts sold to any shop at the published catalogue price; the shop passes them through at list plus a handling margin	60–70% of a visit is material; LLPs 40–60% of parts when replaced, single-source	Air Cargo Week, 2025/26; Safe Fly Aviation, 2026
Escalation	Annual catalogue increases	Spares ~10% a year to late 2023; double-digit 1 Nov 2022; high-single-digit Aug 2023; "above inflation by 3 to 4 points" (Safran); -7B parts +20–30% after the Aug 2023 revision (Magnetic); LLP set ~7% a year 2008–25 (derived); 2024–26 steps not found	Aviation Week, late 2023; Magnetic via Aviation Week, 2024; Part I
Transactional share	Visits bought one at a time, time-and-materials, with the operator or its shop choosing the parts	~85% of CFM56 aftermarket revenue is spares and time-and-materials, against ~40% for the LEAP; ~15% of the CFM56 fleet under long-term agreements, ~50% of the LEAP fleet	Safran via Aviation Week, c. 2023/24; Aviation Week, undated
Long-term service agreement (LTSA)	A multi-year contract at a fixed rate per flight hour under which the manufacturer performs or pays for the visits and chooses the shop and the parts	15% of the CFM56 fleet; LEAP targeted at 60–70% by 2030	Safran via Aviation Week
TrueChoice (GE)	GE's branded menu: Flight Hour (per-hour agreements); Overhaul (visits in GE's shops); Material (new and used OEM parts and repairs sold to third-party shops and airlines); Transitions (green-time leases, exchanges, material buy-back, shorter builds with maximum used material)	GE states "GE and CFM are the largest used serviceable material provider for their products"; CFM56 is "an open network ... with multiple providers who compete"	GE Aerospace releases and statements, 2026

Element	What it is	Binding figure	Source and level
EngineWise (IAE)	Pratt & Whitney's aftermarket brand under which IAE's network performs V2500 visits	FTAI's five-year agreement covers "over 100" V2500 restorations (6 Jun 2024); V2500 visits in 2025: "slightly more than 800" (RTX CFO) against ~1,400 forecast (Aviation Week data tool)	RTX release, 6 Jun 2024; Aviation Week, 2026
Materials agreements	Multi-year supply of OEM parts and repairs to a shop or airline	FTAI–CFM agreement of 22 Jan 2026 "secures OEM replacement part supply, thrust performance upgrades and component repair"; terms undisclosed; Ryanair's covers ~2,000 CFM56 and LEAP engines	FTAI release, 22 Jan 2026; GE 10-K FY2025

Source: Air Cargo Week, "The true cost of engine maintenance", 2025/26; Safe Fly Aviation, 2026 (consultancy blog); Aviation Week, "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023, "CFM Aftermarket Shifting To Long-term Agreements, Safran Says", c. 2023/24, "Leap Aftermarket Poised For Growth", undated, and "Pratt & Whitney Sees High-Single-Digit MRO Growth This Year", 2026; GE Aerospace investor-relations statements, 2026; GE 10-K FY2025; RTX / Pratt & Whitney release, 6 Jun 2024; FTAI / GlobeNewswire, 22 Jan 2026. Compiled from Part IV §28. Reconciliation R-009, R-018. Accessed 2026-10-03.

Put the first three rows together and the manufacturer's capture on this engine is plain. Most visits are chosen and paid for visit by visit; within each, 60 to 70% of the bill is material; most material is bought new at list. So the manufacturer receives the largest slice of nearly every CFM56 visit without performing it and without a contract that obliges the operator to come back, and its annual escalation lands on that slice whoever runs the shop. Under a time-and-materials visit the operator or its shop decides whether a part is new, used, repaired or an approved alternative; under a long-term agreement the manufacturer decides. The substitutes of S8 and the exchanges of S9 therefore address the 85% of the CFM56 aftermarket that is transactional, and would address far less of a fleet contracted the way the LEAP is being contracted. [Part IV §28–§29] The manufacturer is also present on the other side of the substitutes' line: it sells used material and exchanges through TrueChoice, and it supplies FTAI's own shops with parts and repairs under the January 2026 agreement whose terms were not disclosed.

An operator whose engine is at its life limit therefore has four options, and the whole market is visible in them: a full shop visit with new life-limited parts, in which case the price is the manufacturer's price and the wait is the slot plus the turnaround; a module exchange, which S9 sells; a green-time engine leased to fly out the remaining months; or a sale of the engine for its parts and the purchase of another. The second, third and fourth exist only because the engine is modular and because a secondary market in modules, time and used material exists, and low

retirements shrink the supply behind all three at exactly the moment the slot shortage makes the first slow and escalation makes it expensive. [Part IV §30]

Depth: Part IV §25–§30.

S8. The substitutes for new parts: used material from teardowns, and PMA and DER

Three things can replace a new part from the manufacturer in a shop visit: a used part from a dismantled engine, a new part made by someone else under a regulator's approval, and a repair that keeps the old part in service. All three are priced against the manufacturer's list price, and all three depend on a supply or an approval the manufacturer does not control.

USED SERVICEABLE MATERIAL (USM)

Used serviceable material (USM) is any part that has been in service, has been removed, has been inspected and if necessary repaired to the manufacturer's manual, and has been released on an airworthiness tag with its full history, so that it is legally fit to be installed again. It is made by the manufacturer, so it sits outside the debate over alternative parts, but it competes for the same line in the parts bill. A part with a gap in its paperwork is scrap whatever its condition. Used material trades at 60 to 80% of a new part's price on average, up to 85% early in a platform's life, and above list when the manufacturer cannot deliver (Aviation Week); a specialist trader sells repaired engine material to an airline at 70 to 75% of list (Aircraft Commerce). Because the base is the catalogue, the used price rises with every catalogue revision even if the discount does not move. [Aviation Week, undated; Aircraft Commerce Issue 120; Part V §31, §34; reconciliation R-022]

TEARDOWN, PART-OUT AND FEEDSTOCK

Part-out is the owner's decision to dismantle a retired aircraft or engine and sell its components individually; teardown is the physical disassembly, about three months for an airframe. Feedstock is the trade's word for the aircraft and engines bought, or taken on consignment, to be torn down; FTAI's filings extend it to the unserviceable module a customer returns in an exchange and to engine inventory held for module production. A serviceable module from a teardown is valued like an engine in miniature: its parts at used-material discounts to list, plus the remaining life of the life-limited parts inside it (life fraction times list price, discounted), less the repair needed to make it serviceable. Its supply is a lagged function of retirements. [Part V §32–§33, §36; registry C-14]

WORKED EXAMPLE S.8 — WHY ONLY RUN-OUT ENGINES BECOME FEEDSTOCK, AND HOW MANY THERE ARE

Inputs: a run-out CFM56 is bought for \$0.8–1.2M and yields \$1.6–2.2M of used material, a net \$0.4–0.8M (sourced: Safe Fly Aviation, 2026, consultancy blog); a half-life -7B has a market value of

\$5.7M (-7B24) or \$6.4M (-7B27) (sourced: IBA, Sept 2025); installed base about 24,000, about 23,000 or more than 22,800 (sourced as in S5); retirement rate about 1.5% (2025 actual) or about 2% (2026 expected) (sourced: GE via Leeham).

teardown yield as a share of a half-life engine's value: $1.6 \div 6.4 = 25\%$
to $2.2 \div 5.7 = 39\%$
engines leaving the base a year: $24,000 \times 1.5\% = 360$; $\times 2\% = 480$;
23,000: 345 / 460; 22,800: 342 / 456
on the consultancy base, $14,200 \times 2\% = 284$; first-quarter 2026 "below 1%"
annualised on 24,000: fewer than 240

No owner tears down a \$6M engine for \$2M of parts, so engines with life are sold or leased whole and only engines at or near run-out reach the teardown market. The corollary runs the other way: when half-life engines are scarce and expensive, as the appraiser describes 2024 to 2026, owners of part-worn engines lease them for their remaining months rather than tear them down, and the feedstock pool shrinks further. Gross engine retirements on the manufacturer-level bases and GE's realised rate are roughly 340 to 480 a year, of which an unknown share is torn down, since some are sold whole, some leased for green time first, and from 2026 some earmarked as power-generation cores (S11). Limits: one consultancy's ranges; the bases may include spares; the number actually torn down was not captured. [Part V worked examples 5.1 and 5.4; Part II §16]

The 2023 to 2026 shortage follows in six steps that the sources describe. Airlines kept CFM56-powered aircraft flying because new deliveries slipped and new engines spent time off wing. The expected post-2022 retirement wave did not arrive. The engines that would have been feedstock were leased as green-time engines or overhauled instead. Teardown companies competed for a smaller pool. And the price of everything in the pool rose: CFM56 values up "as much as 50%" in two years on one trade-press account, -7B rents about a third above 2019 on both the appraiser's and the consultancy's figures, "a lessors' market" (IBA, April 2024) and "a seller's market for used parts" (Aviation Week), with AerSale's whole-asset sales falling from 20 engines and one aircraft in 2024 to 13 engines in 2025 in what its management called "a hypercompetitive feedstock environment". [ePlaneAI; IBA via AviTrader, 25 Apr 2024; Aviation Week; AerSale, 5 Mar 2026; Part V §33] The manufacturer is also the largest used-material trader by its own statement, buying engines back under TrueChoice Transitions and selling their material through TrueChoice Material, so an independent teardown company competes with the OEM for feedstock and for customers at once. [GE Aerospace, 2026; Part V §33.6] FTAI sits on both sides of this trade: it buys engines and whole aircraft for their material, and it sells the used modules and parts it does not build into modules through an exclusive partnership. The FY2025 annual report does not name the partner; AAR (a parts distributor and maintenance company) announced on 27 March 2025 that the two had extended their exclusive Serviceable Engine Products agreement for CFM56 material through 2030, covering a pool of more than 450 engines as trade coverage

of the release carried it, and the agreement's terms were not read. [FTAI 10-K FY2025; AAR / PR Newswire, 27 Mar 2025, body as carried by ePlaneAI; Part V §35]

PARTS MANUFACTURER APPROVAL (PMA)

A Parts Manufacturer Approval (PMA) is the FAA's combined design-and-production approval that lets a company other than the type-certificate holder legally make and sell a replacement part for a certificated engine. The applicant must show by its own tests and computations that the part meets the airworthiness standards, or that it is identical to the original, with no access to the manufacturer's data. Approvals have historically concentrated in consumables and simple parts; HEICO's Flight Support Group adds "approximately 300 to 500" or "approximately 400 to 550" a year, two readings of one filing, and earned \$750.4M of operating income on \$3,117.3M of sales in its year to 31 October 2025. A cooled, coated high-pressure turbine blade is the hard case, because its life depends on an alloy, a crystal structure and a coating the applicant must reproduce and prove in an engine-level test programme it funds itself. Chromalloy's FAA approval of a CFM56-5B/7B high-pressure turbine blade on 30 October 2025, "the only aftermarket alternative to the OEM manufactured new part", is treated in the trade as that milestone. No PMA exists for any CFM56 life-limited part. [eCFR, 14 CFR (Code of Federal Regulations) Part 21 Subpart K; HEICO 10-K FY2025; Chromalloy release, 30 Oct 2025; Part VI §37–§38, §41; reconciliation R-024]

DESIGNATED ENGINEERING REPRESENTATIVE (DER) REPAIR

A Designated Engineering Representative (DER) is an engineer appointed by the FAA to approve engineering data on its behalf. A DER repair is a major repair whose data a DER approved rather than being taken from the manufacturer's shop manual; the part stays the manufacturer's part, and the repair saves money by restoring what the manual would scrap, so that no new part of any origin is bought, or by restoring it more cheaply than the manual's repair. The cadence decides which lever applies: at a first shop visit, around 20,000 cycles, airlines typically repair the high-pressure turbine blades; at a second, around 30,000, the hot-section parts typically cannot be repaired again and a new part is needed, which is where an approved alternative blade competes with the manufacturer's. [FAA Order 8110.37F; In Practise, undated; Part VI §40]

EXHIBIT S.8 — THE THREE SUBSTITUTES: WHAT EACH REPLACES, HOW IT IS PRICED, WHO SELLS IT, AND WHAT THE RESEARCH COULD AND COULD NOT SOURCE

Substitute	Replaces	Pricing basis	Who sells it	Binding figures	Not retrieved
Used serviceable material	New OEM parts, and life-limited parts with cycles remaining	60–80% of new on average; 70–75% of list from a trader to an airline; above list when the OEM cannot deliver; LLPs at life fraction × list, then discounted	Teardown traders (AerSale, GA Telesis, others); lessors' part-outs; the manufacturer itself; FTAI through its AAR partnership	Run-out CFM56 \$0.8–1.2M → \$1.6–2.2M of material; market size \$5.87B, \$7.7B or \$8.55B for 2025 by three vendors with scopes unstated	Prices of used LLP packages; the number of engines torn down a year
PMA parts	New OEM blades, vanes, seals, consumables; never a disk, spool or shaft	A discount to list that no retrieved source discloses	HEICO (300–500 or 400–550 approvals a year); Chromalloy, with FTAI as partner	HPT blades and vanes are 15–25% of a visit's parts cost; Chromalloy's CFM56 roster is three announced approvals by one reading and four parts by another	The discount; the saving per visit; the JV's terms
DER repair	A new part, by keeping the old one in service	Repair price against the new part's list	Turbine-part repair houses; airline engineering; independent shops; the manufacturer, which says it has industrialised about 400 CFM56 repairs since early 2023	Repairs are 10–20% of a visit	A repair price for any CFM56 hot-section part

Substitute	Replaces	Pricing basis	Who sells it	Binding figures	Not retrieved
FTAI–Chromalloy joint venture	Hot-section parts in FTAI's own modules	"At cost", per investor blogs; not in filings	FTAI "develops and manufactures PMA parts through a joint venture" (10-K)	Blogs claim "up to \$2 million per shop visit" and 5–10 margin points; FTAI's only quantified lever is "up to \$75,000" a visit from its Bristol accessories venture; an "Advanced Engine Repair JV" (25%, \$22.4M) of unknown identity sits in the equity-method note	Partner, ownership and accounting in the filings; the discount; the saving per module

Source: Aviation Week, "MRO Memo: A Seller's Market For Used Parts", undated; Aircraft Commerce Issue 120; Safe Fly Aviation, 2026 (consultancy blog); The Business Research Company, Market.us and Fortune Business Insights (market-research vendors); HEICO 10-K FY2025; Chromalloy release, 30 Oct 2025; CFM International, 2025/26; FTAI 10-K FY2025; Insider Monkey, FTAI Q3 2025 call transcript; Komodo Capital and Kairos Research (investor blogs). Compiled from Parts V and VI. Reconciliation R-021 to R-027. Accessed 2026-10-03.

Two things about the alternative-parts lever are fixed by the arithmetic of S2 and should travel with the reader. It touches only the on-condition parts line: life-limited parts, 40 to 60% of parts cost when replaced, remain single-source at the escalating list price, and labour, test and the shop's margin are untouched. And it applies only at visits where blades are scrapped rather than repaired, which the interview places at second and later visits, exactly the visits a fleet kept past its planned retirement undergoes. [Part VI §44] The claimed saving is contested at three source levels. Investor blogs put the FTAI–Chromalloy venture's five parts at "up to \$2 million per shop visit"; priced per blade from an interview a set of 80 blades is \$1.6M, which cannot reach \$2M on its own; priced from the consultancy's share the blades and vanes together are \$0.11 to \$0.63M on either visit range; and the one saving FTAI itself has quantified is \$75,000 a visit from a different venture. The research shows what would have to be true and stops. [Komodo Capital; Kairos Research; In Practise; Safe Fly Aviation, 2026; Insider Monkey, Oct 2025; reconciliation R-015, R-026, R-027] On the manufacturer's side, CFM's TRUEngine designation and lessors' return conditions that require only the manufacturer's parts are the levers against alternative parts, and an appraiser records that approved alternatives "continue to undermine full-life maintenance adjustments". [CFM International; Aircraft Value News, undated; Part VI §42]

Depth: Part V §31–§36; Part VI §37–§44.

S9. FTAI's answer: the Module Factory and the Aerospace Products segment

FTAI's module business is the fourth lever of S2 organised as a production line. It does the disassembly, inspection, repair and rebuilding before any customer engine arrives, holds the finished modules in inventory, and sells them into engines whose one expired module would otherwise send the whole engine through a 90-to-120-day visit after a two-to-six-month wait.

MODULE FACTORY

The Module Factory is FTAI's name for its CFM56-7B and -5B modular repair and refurbishment programme, in the FY2025 annual report's words "a dedicated commercial maintenance program designed to focus on modular and parts repair and refurbishment"; the language first used in 2022 and still in the filing places it at a 500,000-square-foot Montréal facility "with a capacity for up to 300 shop visits per year". It is a place (Montréal, bought outright for \$170.0M on 9 September 2024), a programme, and a process: a production line that builds and holds an inventory of serviceable modules for exchange. Material enters it from customer cores returned in exchanges, from engines retired from FTAI's own leasing fleet, from the capital vehicles' aircraft, and from engines and aircraft bought for the purpose, such as the ten off-lease 737-700s acquired from WestJet in September 2026. [FTAI 10-K FY2025; FTAI releases, 9 Sep 2024 and 28 Sep 2026; Part VII §46]

MODULE EXCHANGE

A module exchange separates a customer's engine at the major-module flanges, removes the module that triggered the removal, installs a serviceable module from inventory of compatible build standard and documented life-limited-part status, tests the engine and returns it. The customer pays an exchange fee and gives up the unserviceable module, the returned core, which becomes FTAI's feedstock for a later serviceable module. The industry prices an exchange as the fee plus the difference in life-limited-part life between the module handed over and the module received, adjusted for the condition of the other hardware. Under US revenue rules the returned core is noncash consideration measured at fair value, so FTAI books revenue equal to the fee plus the core's fair value, receives the fee in cash and a module in kind, and carries the module in inventory until it is sold again. No provider's exchange fee, FTAI's included, was found in any source. StandardAero lists module changes among its quick-turn services and GE sells exchanges under TrueChoice Transitions. [Part VII §46–§47; reconciliation cluster K-01]

MAINTENANCE, REPAIR AND EXCHANGE (MRE), AND MRE CONTRACT REVENUE

MRE is FTAI's name for its offer, chosen as a contrast with the industry's MRO: instead of restoring a customer's engine while the customer waits, FTAI swaps in a serviceable module or engine and takes the unserviceable unit as feedstock. It is sold as engine, module and parts sales to airlines, lessors and other shops on the "Aerospace products revenue" line (\$455.0M in 2023, \$1,079.8M in 2024, \$1,600.5M in 2025), which no retrieved filing splits into engines, modules, parts and repairs. MRE Contract revenue is a second line, new in 2025, for sales to the special-purpose vehicles of the 2025 Partnership that FTAI manages and part-owns: \$335.8M for 2025, \$221.2M in the first quarter of 2026, \$182.8M in the second and \$404.0M for the half, against \$170.2M a year earlier. Two filings define it differently: the Q3 2025 10-Q describes "stand-ready obligations to provide replacement CFM56-5B and CFM56-7B engines to customers as they become unserviceable", and the Q2 2026 10-Q describes "the transaction price for sales of CFM56-5B, CFM56-7B and V2500 engines and related modules to the SPVs of the first SCI partnership"; one is an obligation and the other a sale, and both are carried. [FTAI 10-K FY2025; FTAI 10-Qs Q3 2025 and Q2 2026; Part VII §45, §48; reconciliation R-044]

EXHIBIT S.9 — AEROSPACE PRODUCTS BY PERIOD: REVENUE, THE RELATED-PARTY LINE, ADJUSTED EBITDA, MARGIN AND MODULES

Period	Aerospace products revenue	MRE Contract revenue	Segment revenue	Segment Adjusted EBITDA	Margin on segment revenue	Modules produced	Source
FY2022	\$85.1M	n/a	\$85.1M	\$27.4M	32.2% (derived)	not retrieved	FY2022 10-K
FY2023	\$455.0M	\$0	\$455.0M	\$160.0M	35.2% (derived)	not retrieved	FY2025 10-K
FY2024	\$1,079.8M	\$0	\$1,079.8M	\$380.6M	35.2% (derived)	Q4 ≈136 (derived)	FY2025 10-K; Q4 2025 call
FY2025	\$1,600.5M	\$335.8M	\$1,936.2M	\$671.3M	34.7% (derived); Q4 2025 stated 35%	757 (target 750); Q2 184; Q4 228	FY2025 10-K; releases; call coverage
Q1 2026	\$522.6M	\$221.2M	\$743.8M	\$222.6M	29.9% (derived)	≈270 (derived)	Q1 2026 10-Q and release

Period	Aerospace products revenue	MRE Contract revenue	Segment revenue	Segment Adjusted EBITDA	Margin on segment revenue	Modules produced	Source
Q2 2026	\$692.2M (derived)	\$182.8M	\$875.0M	\$249.7M	28.5% (derived)	296	Q2 2026 10-Q and release
1H 2026	\$1,214.8M	\$404.0M	≈\$1,618.8M (derived)	≈\$472.3M (derived)	≈29.2% (derived)	566	Q2 2026 10-Q; releases
2026 guidance				\$1,050M (25 Feb 2026, reaffirmed 29 Jul 2026)	"can approach 40%" (26 Feb 2026 call)	1,050 (Feb 2026 release); raised to 1,200 on the July 2026 call (call coverage)	Releases; MarketBeat; GuruFocus
2027 guidance				\$1,400M (29 Jul 2026)			Q2 2026 release

Source: FTAI Aviation, Form 10-K for FY2025 (segment table and segment geographic table) and Form 10-K for FY2022; Forms 10-Q for Q1 and Q2 2026; earnings releases of 29 Jul 2025, 25 Feb 2026, 29 Apr 2026 and 29 Jul 2026; Motley Fool and MarketBeat coverage of the Q4 2025 call, 26 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat coverage of the Q2 2026 call, 30 Jul 2026. Derived figures are arithmetic on filed numbers, done in the research. Compiled from Part VII Exhibits 7.4, 7.5, 7.11, 7.12 and 7.18. Reconciliation B-1, R-036, R-043. Accessed 2026-10-03.

Four tensions sit in that table and the primer presents each both-sided. Segment revenue equals the products line plus the MRE line: the 10-K's geographic table totals \$1,936,244 thousand, which is exactly \$1,600,456 thousand plus \$335,788 thousand, and the 2026 releases agree; the explicit segment-note sentence mapping the MRE line to the segment was not retrieved, and one dossier treated the placement as open. [reconciliation B-1, R-043] The 2026 module target has two figures: 1,050 was set with the fourth-quarter 2025 results on 25 February 2026, a 39% increase on 757, and coverage of the 30 July 2026 call reported it raised to 1,200; the July release text was not found to carry 1,200, and one summary of the same call still read 1,050. [FTAI release, 25 Feb 2026; GuruFocus, Nasdaq.com and MarketBeat, 30 Jul 2026; reconciliation R-036] Cost of sales for 2025 has two figures in one filing: \$1,240,368 thousand is the segment's own cost of sales in the segment expenses table, and \$1,349,719 thousand is the consolidated line, which also carries the carrying value of aircraft and engines the Leasing segment sells; the segment figure gives a derived gross margin of 35.9%, within about a point of the 34.7% Adjusted EBITDA margin. [FTAI 10-K FY2025;

reconciliation R-045] And the margin moved from 35% in the fourth quarter of 2025 to about 29 to 30% in the first half of 2026, the period in which the related-party share of revenue was highest (17.3% for 2025; 29.7%, 20.9% and 25.0% for the first quarter, the second and the half); the research recorded the two movements side by side and declined to infer a cause, because the filings disclose neither margin by revenue line nor the composition of the profit elimination that S10 describes. [Part VII §48, §53]

Capacity is stated in four units at four dates for different footprints, and the figures measure different things rather than contradicting one another: "up to 300 shop visits per year" for Montréal as a conventional shop (10-K, 2022 language); 1,350 module overhauls and more than 500 engine tests a year for Montréal plus Miami (release, 9 Sep 2024); 1,800 modules, 600 engines and more than 600 tests after Rome Fiumicino was added (release, 26 Feb 2025); and 3,000 modules a year across all sites and partners, including Lisbon, Orange, Bristol, Indonesia and Egypt, "supporting its 25% market share goal and 100 Mod-1 units annually" (call coverage, July 2026). FTAI's own releases convert at three modules per engine, which matches the gas-path count and the ratio of its module to test figures; Part I counts four major modules including the gearbox, so 1,200 modules is 400 engines on FTAI's convention and 300 on the other. Lisbon's "113,000 sq ft, 300+ modules" is call coverage, unverified. [FTAI 10-K FY2025; FTAI releases, 9 Sep 2024 and 26 Feb 2025; GuruFocus and Quartr, Jul 2026; reconciliation R-035, R-037, R-041] The sites "span over one million square feet"; 985 full-time employees and independent contractors worked for the company at 31 December 2025, of whom 494 were full-time employees in Canada, where Montréal is, about 71% of them under collective agreements. [FTAI 10-K FY2025]

WORKED EXAMPLE S.9 — REVENUE AND ADJUSTED EBITDA PER MODULE: A BOUND, NOT A PRICE

Inputs: Aerospace products revenue of \$1,600.5M (FY2025), \$522.6M (Q1 2026) and \$692.2M (Q2 2026, derived); segment revenue including MRE of \$1,936.2M, \$743.8M and \$875.0M; segment Adjusted EBITDA of \$671.3M, \$222.6M and \$249.7M; modules produced of 757, about 270 (derived) and 296 (all sourced as in Exhibit S.9); 2026 segment guidance of \$1,050M over targets of 1,050 and 1,200 modules.

products revenue per module: FY2025 $1,600.5 \div 757 = \$2.11\text{M}$; Q1 2026 $\$1.94\text{M}$; Q2 2026 $692.2 \div 296 = \$2.34\text{M}$

segment revenue per module: FY2025 $\$2.56\text{M}$; Q2 2026 $875.0 \div 296 = \$2.96\text{M}$

Adjusted EBITDA per module: FY2025 $671.3 \div 757 = \$0.89\text{M}$; Q1 2026 $\$0.82\text{M}$; Q2 2026 $\$0.84\text{M}$

2026 guidance per target module: $1,050 \div 1,050 = \$1.00\text{M}$; $1,050 \div 1,200 = \$0.875\text{M}$

second half of 2026 needed for 1,050 modules: $1,050 - 566 = 484$, or 242 a quarter against 296 in Q2; for 1,200: 634, or 317 a quarter

For every CFM56 module FTAI produced, the segment booked about \$2M of product revenue and earned about \$0.8 to \$0.9M of Adjusted EBITDA; including the related-party line the revenue figure is \$2.6 to \$3.0M. Five limits make every one of these an upper bound rather than a price: the numerator includes whole-engine sales, parts, used material and V2500 activity while the denominator counts only CFM56 modules; a module produced is not a module sold, and may go into inventory, an FTAI leasing engine or a Mod-1 generator set; segment Adjusted EBITDA includes the profit on related-party sales before its elimination; exchange revenue includes the fair value of returned cores, which is not cash; and the mix between a core module and a fan module, worth multiples apart, is undisclosed. [Part VII §56; reconciliation R-046]

The segment's share of its market is framed three ways, each needing a different denominator: about 5% of a \$22B addressable market as of 2024 (release, 26 Feb 2025); a goal of 25% of CFM56 shop visits, with no year and no denominator stated (call, July 2026); and, derived, about 17% of a 2,350-visit year at 1,200 modules and three modules per engine, which is a material-throughput share and not a share of visits. [FTAI release, 26 Feb 2025; GuruFocus, Jul 2026; reconciliation R-042] Management named three levers for moving the margin from the mid-30s toward 40% in 2026: the approved alternative high-pressure turbine blade, more used material, and the CFM materials agreement, all on the material line of S2. The first half of 2026 ran at about 29%. [MarketBeat and Motley Fool, 26 Feb 2026; Part VII §52, §57]

THE MUDDY WATERS SHORT REPORT AND THE AUDIT COMMITTEE REVIEW

Muddy Waters Research, a short seller, published a report on FTAI on 15 January 2025. As third-party summaries and the later securities complaint describe it, the report alleged that one-time engine sales were reported as maintenance revenue, that whole-engine sales were presented as module sales, that margins were inflated, that sales were pushed to intermediaries ahead of demand, and that depreciation on engines not on lease lowered reported cost of sales; the report itself was not read in the research. FTAI said it "strongly disagrees". On 18 January 2025 its Audit Committee "determined to begin a review, which included engaging independent advisors", and in February 2025, on or about 19 February by trade coverage, announced that "the allegations made against the Company are without merit"; the Q3 2025 10-Q and the FY2025 10-K describe the same exercise as an "internal investigation" by "outside counsel and forensic accountants". A securities class action, S.D.N.Y. No. 25-cv-00541, covers a class period of 23 July 2024 to 21 January 2025; the defendants' motion to dismiss the amended complaint, filed 20 November 2025, was pending at the research date. The Audit Committee engaged a new auditor, KPMG (replacing Ernst & Young LLP, which had served from 2016), effective 17 June 2025; which firm signed the FY2025 report was not confirmed. [FTAI 8-Ks, Jan and Feb 2025; FTAI 10-Q Q3 2025; FTAI 10-K FY2025; Benzinga, 15 Jan 2025; law-firm case pages; Part VII §54; reconciliation R-063, R-064, R-066]

The definitional point behind the allegations is one the reader can now state. A whole CFM56 is three modules in FTAI's convention. Three modules built into an engine and sold outright is an engine sale; the same three installed one at a time into three customers' engines are three exchanges. The revenue booked may be similar, and the filings present engines, modules and parts on one line without disaggregating it, so a reader cannot tell from the filings alone how much is whole engines. [Part VII §47.4] That is the boundary the dispute concerns, and nothing in the primer resolves it.

Depth: Part VII §45–§57.

S10. FTAI's capital: Aviation Leasing, the Strategic Capital vehicles, the money that flows between them, and the 2026 securitization

The other half of FTAI owns things. Aviation Leasing held 290 assets at 31 December 2025, 47 aircraft and 243 engines, including 8 aircraft and 17 engines in Russia; 37 aircraft and 143 engines were on lease, and utilization was about 77% on the company's definition, which weights each asset by its equity value and excludes airframes. Its 2025 lines were lease income of \$235.2M, maintenance revenue of \$218.5M and asset sales revenue of \$106.9M, and its segment Adjusted EBITDA was \$608.9M, so rent cannot account for the earnings figure on its own; maintenance revenue, asset-sale margins, a \$46.4M gain on sales to the first capital vehicle and the equity-method share must together account for most of it, and the segment note that

would show the composition was not retrieved. [FTAI 10-K FY2025; Part VIII §58] The aircraft is bought for the two engines under its wings (S4), so the leasing book is both a source of rent and a feedstock pipeline for S9.

In 2025 the company began selling that book into vehicles funded by other people's money while keeping three roles in them.

STRATEGIC CAPITAL INITIATIVE (SCI): THE 2025 PARTNERSHIP, THE 2026 SPV AND THE SEED ASSETS

The Strategic Capital Initiative (SCI), launched 30 December 2024, is FTAI's asset-management arm. It raises third-party capital into closed-end vehicles that buy on-lease mid-life 737NG and A320ceo aircraft; FTAI manages each vehicle as Servicer, co-invests as a limited partner, and supplies every engine shop visit for the fleet through module and engine exchanges with its MRE business. The 2025 Partnership, the first vehicle, closed on 27 October 2025 at a \$2.0B equity hard cap against a \$1.5B target, carried a \$2.5B asset-level debt commitment arranged in February 2025, and by 17 August 2026 had "approximately \$6.0 billion of total capital across over 300 aircraft", was fully committed, had paid a first distribution on 30 June 2026 and was in "harvest phase"; a release of 22 May 2026 said it then owned 292 aircraft. The Seed Assets are the 46 on-lease aircraft FTAI sold from its own book to start it, "at an estimated net purchase price of \$549 million"; the sales cut leasing equipment at cost from \$2,963.5M to \$2,057.6M during 2025. The 2026 SPV, the second vehicle, closed a \$2.0B warehouse facility from 13 lenders with an accordion to \$3.0B on 14 August 2026; its equity size and FTAI's stake were undisclosed. The release's "\$5.5 billion of warehouse financing" matches the disclosed \$2.5B plus \$2.0B only if the accordion is counted. [FTAI releases, 30 Dec 2024, 26 Feb 2025, 27 Oct 2025, 22 May 2026, 17 Aug 2026; FTAI 10-K FY2025; Part VIII §60–§62; reconciliation R-055]

SERVICER AND SERVICING FEE

The Servicer is the party contracted to manage a vehicle's assets it does not wholly own: leasing and remarketing the aircraft, collecting rent and reserves, overseeing their technical condition and selling them, for a fee. FTAI "as the Servicer, provides aircraft management services to the 2025 Partnership, and receives customary, market-based compensation": \$10.2M for 2025, \$6.99M for the second quarter of 2026 and \$12.85M for the first half. The fee's rate, its base (committed equity, invested cost or rents) and the income-statement line it sits on are not disclosed. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; Part VIII §61, §63]

EQUITY METHOD AND PROFIT ELIMINATION

Under the equity method an investor with significant influence but not control carries its stake at cost plus its share of the investee's earnings, less distributions and eliminated profit, on one balance-sheet line and one income line, with none of the investee's revenue, assets or debt consolidated. FTAI's limited-partner interest in the 2025 Partnership is described as 19% in the FY2025 10-K and the 2026 10-Qs and as 20% in the Q3 2025 10-Q; it is accounted for this way and carried at \$281.7M (31 Dec 2025), \$279.4M (31 Mar 2026) and \$365.5M (30 Jun 2026). Profit elimination removes from current earnings the profit on sales to such an investee and recognises it later: "the profit from the MRE Contract revenue is eliminated through equity method earnings and will be recognized over time as the 2025 Partnership generates income". Eliminations were \$10.0M in the first quarter of 2026 and \$6.6M in the second, \$16.6M for the half. Whether the eliminated amount is 19% of the profit or all of it is not stated. [FTAI 10-Qs Q3 2025, Q1 and Q2 2026; FTAI 10-K FY2025; Part VIII §63–§64; reconciliation R-047, R-048]

ASSET-LIGHT MODEL AND HARVEST PHASE

"Asset-light" is management's term for owning fewer leased assets directly and earning instead servicing fees, maintenance work and equity income from vehicles that own them; the July 2026 cut to Aviation Leasing guidance was explained as "reflecting a continued shift to an asset-light business model". The balance sheet also carries the 19% interest at \$365.5M, about \$380M at full commitment. Harvest phase is the period after a closed-end vehicle has finished buying, in which it collects rent, pays distributions and sells assets until it winds up; the 2025 Partnership was described as in harvest phase on 17 August 2026, and its term and exit route were not located. [FTAI releases, 29 Jul 2026 and 17 Aug 2026; FTAI 10-Q Q2 2026; Part VIII §60–§61; reconciliation R-058]

EXHIBIT S.10 — THE STREAMS BETWEEN FTAI AND THE 2025 PARTNERSHIP, BY PERIOD, AND WHERE EACH LANDS

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands in FTAI's accounts	Source
Gain on aircraft sold to the Partnership	\$46.4M (the three quarters of 2025 sum to \$50.1M on 45 aircraft)	n/l	n/l	n/l	Gain line outside revenue, under topic 610-20 of the US Accounting Standards Codification (ASC), described as "non-recurring"; inside Aviation Leasing Adjusted EBITDA	10-K FY2025; 10-Qs 2025

Stream	FY2025	Q1 2026	Q2 2026	1H 2026	Where it lands in FTAI's accounts	Source
Servicing fees	\$10.2M	\$5.9M (derived)	\$7.0M	\$12.85M	Line not confirmed; presumed other revenue	10-K FY2025; 10-Q Q2 2026
MRE Contract revenue, sales to and from the Partnership	\$335.8M	\$221.2M	\$182.8M	\$404.0M	MRE Contract revenue line, Aerospace Products (S9)	10-K FY2025; 10-Qs 2026
Profit elimination on those sales	n/l	\$10.0M	\$6.6M	\$16.6M	Through equity-method earnings	10-Qs 2026
Equity-method income	n/l	n/l	n/l	n/l	Equity in earnings of unconsolidated entities	not located
Carrying value of the stake; stake	\$281.7M; 19% (20% in Q3 2025)	\$279.4M; 19%	\$365.5M; 19%		Equity-method investment, balance sheet	10-K FY2025; 10-Qs
Aviation Leasing guidance for 2026	\$575M (25 Feb 2026)		\$475M (29 Jul 2026)		Segment Adjusted EBITDA; 2027 guidance \$450M	Releases

Source: FTAI Aviation, Form 10-K for FY2025; Forms 10-Q for the quarters ended 31 Mar 2025, 30 Jun 2025, 30 Sep 2025, 31 Mar 2026 and 30 Jun 2026; releases of 25 Feb 2026 and 29 Jul 2026. n/l = not located in anything the research reached. Compiled from Part VIII Exhibits 8.6, 8.10 and 8.11. Reconciliation R-047, R-048, R-049, R-054, R-057. Accessed 2026-10-03.

Three disagreements in that table bear directly on the economics and the primer presents each both-sided. The stake is 19% in the FY2025 10-K and the 2026 10-Qs and 20% in the Q3 2025 10-Q; the notes suggest dilution at the final close and mark that as a guess. [reconciliation R-048] The 2026 Leasing guidance was \$575M on 25 February 2026 and \$475M on 29 July 2026; one search reading of the July release returned \$575M, which the orchestrator's follow-up established as a mis-read of the February figure carried forward, so the path is \$575M cut to \$475M, and 2027 is \$450M. [FTAI releases, 25 Feb 2026 and 29 Jul 2026; reconciliation R-049] And the profit elimination supports two readings of one sentence. If FTAI eliminates only the share of profit corresponding to its stake, as is usual for sales to an equity-method investee because the other owners paid an

arm's-length price, then \$16.6M of eliminations implies about \$87M of gross profit on \$404.0M of first-half sales to the Partnership, a margin of about 22% (at 20%, \$83.0M). If it defers all of the profit, the gross profit was \$16.6M and the margin about 4%. The filing supports either, and a reader comparing the segment's margin on Partnership volume with its margin on third-party volume cannot do so from the filings. [FTAI 10-Q Q2 2026; reconciliation R-047] Two further facts are carried as the filing states them: the aircraft sales into the Partnership are booked as "non-recurring in nature and not considered part of the Company's ordinary activities", while the launch release framed "\$3.0+ billion of capital annually" and the Q2 2026 10-Q says FTAI's primary investment activity will run through the initiative going forward; and the related-party pricing of aircraft, of exchanges and of the fee is not disclosed, the agreements having been listed as exhibits to the Q2 2026 10-Q and not read. [FTAI 10-K FY2025; FTAI 10-Q Q2 2026; reconciliation R-057]

WORKED EXAMPLE S.10 — WHAT A \$6.0B VEHICLE PAYS ITS MANAGER: THE DISCLOSED RUN-RATE AGAINST THREE PEER SCALES

Inputs: servicing fees of \$6.988M for Q2 2026 and \$12.849M for 1H 2026 (sourced: 10-Q Q2 2026); total capital about \$6.0B and equity \$2.0B (sourced: releases, 17 Aug 2026 and 27 Oct 2025). For the rent-based scales, a lease rate factor of 0.9% a month (assumed in the research; the derived alternatives are 0.70%, 0.78% and 1.0%, S4) on \$6.0B, and the peer fee scales disclosed in their filings: Carlyle's aviation securitizations at about 2% of rents plus incentive fees of up to 5% of rents and 3% of sale proceeds; Willis Lease's engine trusts at 11.5% plus 2.0% of net rents and 3% of sale proceeds; a private-fund convention of 1.5% of committed equity (assumed, not an SCI term).

disclosed run-rate: $6.988 \times 4 = \$27.95\text{M}$ a year; $12.849 \times 2 = \$25.70\text{M}$
on total capital: $27.95 \div 6,000 = 0.47\%$; on equity: $27.95 \div 2,000 = 1.40\%$

rent at 0.9% a month on \$6.0B: $0.009 \times 12 \times 6,000 = \648M a year (at
0.70%: \$504M; at 1.0%: \$720M)

Carlyle scale: servicing 2% of rents = \$13.0M; incentive at the 5% cap =
\$32.4M; sales fee 3% on an assumed \$600M of disposals = \$18.0M

private-fund scale: $1.5\% \times 2,000 = \$30.0\text{M}$ a year

Willis trust scale, for contrast only: $13.5\% \times 648 = \$87.5\text{M}$

The disclosed run-rate of \$26 to \$28M sits between the servicing-only securitization figure of about \$13M and the private-fund management fee of \$30M, and well below the Willis scale, which is a fee inside a consolidated vehicle. That placement depends on an assumed rent factor and assumed fee conventions, and the filings disclose neither the rate nor the base of FTAI's fee; the Partnership's agreement allocates FTAI's share of earnings "based on the contractual profit-sharing arrangement", and whether it contains a hurdle or a promote is undisclosed. Limits: every peer figure is from a different structure; the Q2 2026 quarter is the first after full commitment, so the run-rate may still be

moving. [FTAI 10-Q Q2 2026; Carlyle 10-K FY2025; Willis Lease 10-Ks FY2019 and FY2020; Part VIII worked example 8.5; reconciliation R-030]

The maintenance stream is the one the releases describe but never size. A 300-aircraft fleet carries 600 engines; on the sourced intervals of S1 that is 40 to 100 removals a year, and at the per-module bounds of S9 that is a few hundred million dollars of exchanges a year. The disclosed sales to the Partnership, \$404.0M in the first half of 2026 alone, sit above every case built that way, and the record offers four candidates without choosing: the line covers purchases from the Partnership as well as sales to it; aircraft bought mid-life may have arrived with engines already due; it may include whole engines sold to equip the fleet; and the per-module figures are bounds. The counts of engines and modules behind the dollars were not disclosed. [Part VIII worked example 8.6] The vehicle's own borrowing is in three layers that never reach FTAI's balance sheet: the February 2025 asset-level commitment, the \$612M securitization of May 2026 on 48 of its aircraft, and whatever bank debt remains on the rest; FTAI's own \$3.45B of long-term debt is its senior notes (S12), and whether FTAI was required to consolidate the Partnership was not located. [Part VIII §61, §64] The WestJet transaction of 28 September 2026 shows the structure working in one deal: 17 737-700s on lease went to the 2026 SPV in a sale-leaseback, and 10 off-lease aircraft went to FTAI for their CFM56-7B engines and modules, 20 engines or 60 module-equivalents; price and terms were not disclosed. [FTAI release, 28 Sep 2026; Part VIII §67]

Depth: Part VIII §58–§67; Part X §82.

S11. FTAI Power: a second life for cores, the J&F order, the market it sells into

The newest business removes CFM56 engines from aviation for good. It sits on the same side of the ledger as retirement and teardown, and it competes with both for the same run-out cores.

FTAI POWER: MOD-1, HEAT RATE, BEHIND-THE-METER BRIDGING POWER, AND J&F POWER SYSTEMS

FTAI Power, launched 30 December 2025, converts CFM56 engines into aeroderivative generator sets: the fan is removed and replaced by a small compressor, a power turbine and generator are fitted, and the fuel system is changed for gas. Mod-1 is its 25 megawatt (MW) trailer-mounted unit, described in call coverage as 35 to 40% efficient at a heat rate of about 9,000, deployable in about two weeks, and giving the engine a 10-to-20-year second life on "near-fully depreciated" input engines; the release-level statements are capacity for "over 100 units annually", first delivery by the fourth quarter of 2026 and 100 units in 2027. Heat rate is the fuel energy needed to make one kilowatt-hour, so 9,000 is 37.9% efficient and a 50%-efficient piston engine is about 6,824. Behind-the-meter (BTM) generation is power made on the customer's side of the utility meter, and bridging power is temporary on-site generation used until a grid connection arrives, "five to seven years" on one supplier's description of its customers. J&F Power Systems is the venture with Jereh Group, a Chinese maker of trailer-mounted oilfield gas-turbine packages, through which Mod-1 is packaged and sold: FTAI calls it "FTAI's joint venture"; Jereh's exchange disclosure, as reported, calls it "its subsidiary"; the ownership split and the consolidating party are undisclosed. On 22 July 2026 J&F received a five-year master supply agreement and an initial \$1.465 billion purchase order from "a leading international cloud service provider" (FTAI's release) or "a U.S. hyperscaler" (call summaries), for delivery in batches through November 2027 on milestone payments with a performance adjustment capped at 10%; the unit count was not disclosed. [FTAI releases, 30 Dec 2025 and 22 Jul 2026; FTAI 10-Q Q2 2026; TheValueist and Aviation Week, call coverage, 26 Feb 2026; FilingReader, 22 Jul 2026; Part IX §68–§71; reconciliation R-061]

The market it sells into is defined by a shortage the sources date precisely. The International Energy Agency expects global data-center electricity use to more than double to about 945 TWh by 2030. GE Vernova's gas-turbine backlog plus slot reservations rose from 50 GW in 2025 to 116 GW at the second quarter of 2026, with slots being sold for 2031 delivery; Siemens Energy quotes lead times of three years and more; Wood Mackenzie counted 110 GW of orders at end-2025 against 60 to 70 GW a year of manufacturing capacity. [IEA, Apr 2025; Power Engineering, Jul 2026; Oilprice, 2026; Part IX §71] A buyer that needs power in 2026 or 2027 cannot get a new frame turbine, and that is the opening for a 25 MW unit built from a used airline engine, for piston plants and for fuel cells. The 2027 guidance puts FTAI Power's Adjusted EBITDA at \$450M, confirmed in the release text; one call summary gives "\$450 million to \$750 million", and both are carried. [FTAI release, 29 Jul 2026; GuruFocus, 30 Jul 2026; reconciliation R-062]

WORKED EXAMPLE S.11 — THE IMPLIED PRICE OF A MOD-1, AND THE CFM56 CORES A HUNDRED OF THEM CONSUME

Inputs: order value \$1.465 billion (sourced: FTAI release, 22 Jul 2026); 2027 target 100 units (sourced: release, 25 Feb 2026); unit count in the order undisclosed, so 100, 80 and 60 are assumed; 25 MW per unit; one gas generator per unit (assumed, by analogy with the LM2500 (GE's

aeroderivative of the same class)); installed base and retirement rates as in S5; three modules per engine (FTAI's convention).

at 100 units: $1,465 \div 100 = \$14.65\text{M a unit} = \586 per kW ; at 80: \$18.3M, \$733/kW; at 60: \$24.4M, \$977/kW
 2027 Power Adjusted EBITDA per unit at 100 units: $450 \div 100 = \$4.5\text{M}$ (at \$750M: \$7.5M)
 cores a year at 100 units: 100 engines = 0.42% of 24,000
 share of retirements, OEM-level bases: $100 \div 480 = 21\%$ (2%) to $100 \div 342 = 29\%$ (1.5%); at "below 1%" sustained, about 42–44%
 share on the consultancy base, $14,200 \times 2\% = 284$: 35%
 module-equivalents: $100 \times 3 = 300$ a year = 29% of the 1,050 target, 25% of 1,200, 40% of 2025's 757, 10% of the 3,000 capacity

The implied price sits below the \$2,000 to \$2,500 per kW of current plant-level filings and near Wood Mackenzie's \$600 per kW turbine-only projection for end-2027, on scopes that differ. Every line scales with the undisclosed unit count, and the margin line holds only if FTAI consolidates J&F and recognises the full order value as revenue, which is undisclosed. The cores line is the one that joins this business to the rest of the chain: a hundred engines a year is under half a percent of the base and a fifth to a third of a year's retirements, so the business competes with the teardown trade and the exchange business for the same run-out engines, and if a unit uses only the core module it releases a fan and a low-pressure turbine as feedstock instead of consuming them. Limits: unit count, engines per unit, consolidation and segment reporting are all undisclosed. [Part IX §74; reconciliation R-001, R-004, R-035, R-061, R-062]

EXHIBIT S.11 — MOD-1 AS FTAI DESCRIBES IT: EACH STATEMENT AND ITS SOURCE LEVEL

Statement	Figure	Source level
Platform launched; unit output; annual capacity	30 Dec 2025; 25 MW; "over 100 units annually"	Release; FY2025 10-K
First delivery; 2027 production	By Q4 2026; 100 units in 2027	Releases of 25 Feb and 29 Apr 2026 (8-K exhibits); 10-Q Q2 2026
Efficiency; heat rate; deployment; second life; input cost	35–40%; about 9,000; about two weeks; 10–20 years; "near-fully depreciated" engines	Call coverage of the 26 Feb 2026 call (TheValueist; Aviation Week)
Initial order	\$1.465B through J&F; five-year master agreement; batches through Nov 2027; milestone payments; 10% performance cap; unit count undisclosed	Release, 22 Jul 2026; Jereh disclosure via FilingReader

Statement	Figure	Source level
J&F ownership	"FTAI's joint venture" (FTAI) against "its subsidiary" (Jereh, as reported)	Release and 10-Q; FilingReader (R-061)
2027 Power Adjusted EBITDA	\$450M (release text); "\$450–750M" (one call summary)	Release, 29 Jul 2026; GuruFocus (R-062)
Not disclosed	Variant used; engines per unit; emissions treatment; build location; capital required; whether Power is a reportable segment	Part IX Exhibit 9.4

Source: FTAI Aviation releases of 30 Dec 2025, 25 Feb 2026, 29 Apr 2026, 22 Jul 2026 and 29 Jul 2026; FTAI 10-K FY2025 and 10-Q Q2 2026; TheValueist (X), 26 Feb 2026; Aviation Week, "MRO Memo: FTAI Pitches CFM56 To Power AI Boom", early 2026; FilingReader, 22 Jul 2026; GuruFocus, 30 Jul 2026. Compiled from Part IX Exhibits 9.3 to 9.5. Accessed 2026-10-03.

Depth: Part IX §68–§74.

S12. The company, its numbers and what remains unknown

INTERNALIZATION, AND THE COMPANY IN ONE PARAGRAPH

FTAI Aviation Ltd. is a Cayman Islands company listed on Nasdaq. It began as Fortress Transportation and Infrastructure Investors LLC, listed on the New York Stock Exchange on 15 May 2015 and managed by a Fortress Investment Group subsidiary under an external management agreement, so that it had no employees of its own. It spun off its infrastructure businesses as FTAI Infrastructure on 1 August 2022 and became a Cayman parent on 10 November 2022. Internalization is the termination of an external management agreement, the hiring of the manager's people, and a payment for the fee stream given up; FTAI's closed on 28 May 2024 for a stated \$300.0M, \$150.0M in cash paid on 17 June 2024 and 1,866,949 shares valued at \$150.0M, plus accrued amounts not quantified, so expense lines before and after mid-2024 are not like for like. The Module Factory language dates from 2022, Montréal was bought on 9 September 2024, the capital initiative launched on 30 December 2024, the short report came on 15 January 2025, the first vehicle hit its hard cap on 27 October 2025, FTAI Power launched on 30 December 2025, and the J&F order came on 22 July 2026. [FTAI 10-Qs Q2 and Q3 2024; FTAI 8-K, 28 May 2024; FTAI Infrastructure 10-K FY2024; Part X §75–§77]

The consolidated statement for 2025 shows the chain's money in one place. Revenues were \$2,507.4M: Aerospace products \$1,600.5M, MRE Contract \$335.8M, lease income \$235.2M, maintenance revenue \$218.5M, asset sales \$106.9M and a residual of \$10.5M presumed to be other revenue. Cost of sales was \$1,349.7M, depreciation and amortisation \$225.8M and interest expense \$247.8M. Consolidated Adjusted EBITDA was \$1,190.9M, against \$862.1M in 2024 and \$597.3M in 2023. [FTAI 10-K FY2025; Part X §79–§80] The company reported net income in 2024 and 2025 and consumed cash in its operations in both years: net cash from operations was +

\$129.0M in 2023, −\$188.0M in 2024 and −\$310.7M in 2025. The mechanism is the module business of S9. Cash spent buying assets to sell is an operating outflow, while cash spent buying assets to lease is an investing outflow, so a factory that must hold a pool of serviceable modules ahead of demand shows its purchases as operating cash out until the modules are sold; inventory stood at \$1,364.3M at 31 March 2026, about a year of 2025's consolidated cost of sales. The working-capital lines that would split the outflow were not retrieved. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; Part X §81]

The balance sheet carries five series of senior unsecured notes, 5.50% due 2028 (about \$1,001M), 7.875% due 2030 (\$497M), 7.00% due 2031 (\$718M), 6.875% due 2032 (\$823M) and 5.875% due 2033 (\$498M), at mixed dates; long-term debt net of \$3,451.1M and cash of \$412.2M at 31 March 2026; an undrawn revolver maturing 22 May 2027; and perpetual preferred shares, of which Series A was redeemed in October 2024, Series C's redemption was announced on 15 May 2026, and Series D at 9.50% (\$65.0M) remains, with Series B's status unresolved in the research. Credit ratings and covenant definitions were not retrieved, so every leverage ratio in the primer is an illustration: about \$3.0B of net debt is 2.6 to 2.7 times an annualised quarter of Adjusted EBITDA. [FTAI 10-K FY2025; FTAI 10-Q Q1 2026; Part X §82] The ordinary dividend rose at four consecutive results: \$0.30 a quarter through the second quarter of 2025, then \$0.35 (27 Oct 2025), \$0.40 (25 Feb 2026), \$0.45 (29 Apr 2026) and \$0.50 (29 Jul 2026), on a share count that has stayed near 102 to 104 million since the internalization; a \$500M repurchase program was announced on 15 September 2026, terms not read. [GlobeNewswire release titles; Part X §83] On 6 March 2026 Nicholas McAleese was appointed Chief Financial Officer and Michael Hazan Chief Accounting Officer, succeeding Eun (Angela) Nam. [FTAI release, 6 Mar 2026]

Guidance is given in segment Adjusted EBITDA, before Corporate and Other. For 2025 it was \$1.1 to \$1.15B on 30 December 2024 (Leasing about \$500M, Aerospace Products \$600 to \$650M), and both segments delivered above the range. For 2026 it moved from \$1.4B, to \$1.525B (\$1.0B Aerospace Products and \$525M Leasing, 27 October 2025), to \$1.625B (\$1.05B and \$575M, 25 February 2026), and back to \$1.525B when Leasing was cut to \$475M on 29 July 2026 with Aerospace Products reaffirmed. For 2027 it is \$2.3B: Aerospace Products \$1.4B, FTAI Power \$450M, Aviation Leasing \$450M. [FTAI releases, 30 Dec 2024, 27 Oct 2025, 25 Feb 2026, 29 Jul 2026; Part X §84] Whether FTAI Power is a third reportable segment in the 2026 quarterly reports was not confirmed; the annual report has two segments and the guidance has three lines.

EXHIBIT S.12 — THE BINDING NUMBERS OF THE CHAIN IN ONE TABLE, EACH AS THE RESEARCH FIXED IT

Quantity	Binding presentation	Source
CFM56 in service	~23,000 (CFM, undated); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); >35,000 delivered; consultancy ~14,200 and >19,000 labelled	CFM; Aviation Business News; Safran; Safe Fly
Life limits, -7B	Fan and booster 30,000; HPC and HPT 20,000; LPT 25,000 engine flight cycles; 1,000–1,500 cycles a year	Mini-packs; Aircraft Value News
Mature intervals	Restoration 10,000–15,000 cycles; LLP overhaul 20,000–25,000	Leeham, Mar 2024
Full -7B LLP set at list	\$1.775M (2008); \$3.4M (2018); "\$4m" (Nov 2019); \$5.7M (Jun 2025, scope not visible); ~7% a year compounded (derived)	Aircraft Commerce; Aircraft Value News; MyAirTrade
Shop-visit cost	Restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M"; overhaul \$3.5–4.2M (all one consultancy blog)	Safe Fly Aviation, 2026
Cost shares	Material 60–70%, labour 20–30%, repairs 10–20%; LLPs 40–60% of parts, HPT blades and vanes 15–25%	Air Cargo Week; Safe Fly
Turnaround and slots	90–120 days overhaul, ~45 light (trade press); ~90 sustaining (GE); slot wait 2–6 months (Bain)	Aviation Business News; GE; Bain
Escalation	Spares ~10% a year 2022–23; inflation plus 3–4 points (Safran)	Aviation Week, late 2023
-7B value and rent	Half-life \$5.7M / \$6.4M (IBA, Sept 2025); green-time \$2.8–3.4M, run-out \$0.8–1.2M (Safe Fly); rent ~\$100,000 (IBA, 2024) and \$42,000–48,000 (Safe Fly, 2026)	IBA; Safe Fly
Used-material price	60–80% of new; 70–75% of list from a trader; above list when the OEM is constrained	Aviation Week; Aircraft Commerce
Retirement rate	GE planning 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%; 2026 ~2%; 1Q 2026 "below 1%"	GE via Aviation Week and Leeham; GE IR page
Shop visits a year	2,300–2,400 in 2026–28, apex 2027–28 (GE); ~2,500 peak 2025–26 (Safran, earlier); just above 2,000 in 2019	Aviation Week
OEM model	~85% of CFM56 aftermarket transactional vs ~40% LEAP; ~15% vs ~50% of fleets under long-term agreements	Safran via Aviation Week

Quantity	Binding presentation	Source
Modules produced	Q2 2025 184; Q4 2025 228; FY2025 757 (target 750); Q1 2026 ≈270; Q2 2026 296; 1H 2026 566; 2026 target 1,050 (Feb release), raised to 1,200 (July call coverage)	FTAI releases and calls
Capacity	300 shop visits (Montréal, 2022 language); 1,350 modules (Sep 2024); 1,800 modules / 600 engines (Feb 2025); 3,000 modules (call coverage, Jul 2026); three modules per engine	FTAI 10-K and releases
Aerospace Products	Revenue \$455.0M / \$1,079.8M / \$1,936.2M (2023–25, incl. MRE \$335.8M in 2025); Adjusted EBITDA \$160.0M / \$380.6M / \$671.3M; Q2 2026 \$875.0M and \$249.7M; cost of sales FY2025 \$1,240.4M (segment) and \$1,349.7M (consolidated)	FTAI 10-K FY2025; Q2 2026 release
MRE Contract revenue	\$335.8M (2025); \$221.2M (Q1 2026); \$182.8M (Q2 2026); \$404.0M (1H 2026)	FTAI 10-K; 10-Qs
Aviation Leasing	290 assets (47 aircraft, 243 engines), ~77% utilization; lease income \$235.2M; Adjusted EBITDA \$500.1M / \$608.9M (2024–25); 2026 guidance \$575M → \$475M	FTAI 10-K; releases
SCI	2025 Partnership \$2.0B equity, \$2.5B debt commitment, ~\$6.0B total capital, 300+ aircraft; seed 46 aircraft ~\$549M; stake 19% / 20%; carrying value \$365.5M (30 Jun 2026); fees \$10.2M (2025), \$12.85M (1H 2026); eliminations \$16.6M (1H 2026); 2026 SPV \$2.0B warehouse, accordion \$3.0B; ABS \$612M on 48 aircraft	FTAI releases, 10-K, 10-Qs
FTAI Power	Launched 30 Dec 2025; Mod-1 25 MW; >100 units a year; first delivery Q4 2026; 100 units 2027; J&F \$1.465B (22 Jul 2026); 2027 Adjusted EBITDA \$450M ("450–750M" in one summary)	FTAI releases; call coverage
Consolidated 2025	Revenues \$2,507.4M; cost of sales \$1,349.7M; D&A \$225.8M; interest \$247.8M; Adjusted EBITDA \$1,190.9M; operating cash flow –\$310.7M; inventory \$1,364.3M (31 Mar 2026); long-term debt \$3,451.1M and cash \$412.2M (31 Mar 2026)	FTAI 10-K FY2025; 10-Q Q1 2026
Guidance	2026 \$1.4B → \$1.525B → \$1.625B → \$1.525B (\$1.05B Aerospace Products, \$475M Leasing); 2027 \$2.3B (\$1.4B / \$450M / \$450M)	FTAI releases
Corporate	Listed 15 May 2015; spin-off 1 Aug 2022; Cayman 10 Nov 2022; internalization 28 May 2024, \$300.0M; short report 15 Jan 2025; review 18 Jan to Feb 2025; dividend \$0.30 → \$0.50; \$500M repurchase 15 Sep 2026; 985 employees and contractors, 494 in Canada	FTAI filings and releases

Source: reconciliation section E.2 and the orchestrator's resolutions, as presented in the sections above; each figure's underlying source is cited where it first appears in this Spine. Accessed 2026-10-03.

The chain, read backwards from this table, is one sentence long. A company that owns engines and aircraft bought for their engines, that rebuilds the modules of an engine no longer in production into exchanges for a fleet that did not retire on schedule, that sells those exchanges in part to vehicles it manages and part-owns, and that has begun turning the run-out cores into generators for a power market short of turbines, is a company whose every line depends on the same three quantities: how many CFM56 engines there are, how fast they leave, and what a new part from their only manufacturer costs. Each of the three is contested in the research, and the ten disagreements below are where the contest sits.

Depth: Part X §75–§85; Part XII §94.

Tensions carried in the Spine

Id	The two (or more) figures	Where in the Spine
R-001 / R-002	CFM56 in service: ~23,000 (CFM); ~24,000 (Aviation Business News, Sept 2025); >22,800 (Safran, end-2025); consultancy ~14,200 (read as all variants or as -7B only) and >19,000 (Safe Fly)	S1 box; S5, Exhibit S.5; S7 worked example; S8, S11
R-004 / R-005	Retirement rate: GE 3–4% → 2–3% → 1.5–2%; 2025 ~1.5%, 2026 ~2% (via LeeHam); 1Q 2026 "below 1%"; aircraft retirements 2024: "fewer than 200" (IBA) against 587 (Cirium), scope not visible	S5, Exhibit S.5, worked example S.5; S8; S11
R-013	Shop-visit cost: restoration \$1.2–1.6M and \$1.8–2.5M; heavy with LLPs \$2.1–2.8M and ">\$3.5M" (one consultancy blog, two pages)	S2, Exhibit S.2, worked example S.2
R-016 / R-017	Full -7B LLP set \$5.7M (MyAirTrade, Jun 2025) or ~\$4M (Aircraft Value News, Nov 2019); \$285 or \$200 a cycle	S2, Exhibit S.2, worked example S.2
R-028 / R-029	-7B rent ~\$100,000/month (IBA, Apr 2024) against \$42,000–48,000 (Safe Fly, 2026, may be green-time); value \$5.7M/\$6.4M "levelled off" (IBA) against green-time \$2.8–3.4M and rising rents (Safe Fly)	S4, Exhibit S.4, worked example S.4; S8
R-035 / R-037	Module count: four major modules (anatomy) against three per engine (FTAI releases); capacity 300 visits / 1,350 / 1,800 / 3,000 in different units, footprints and dates	S1 box; S9; S11
R-036	2026 module target 1,050 (release, 25 Feb 2026) against 1,200 (call coverage, 30 Jul 2026)	S9, Exhibit S.9, worked example S.9

Id	The two (or more) figures	Where in the Spine
R-043 / R-044	Segment revenue = products + MRE by arithmetic, segment-note mapping not retrieved; MRE Contract revenue as "stand-ready obligations" (Q3 2025 10-Q) against "the transaction price for sales" (Q2 2026 10-Q)	S9 box, Exhibit S. 9
R-045	Cost of sales FY2025 \$1,240,368K (segment table) against \$1,349,719K (consolidated line); 35.9% derived	S9; S12
R-047 / R-048 / R-049	Eliminations of \$16.6M imply ~\$87M of gross profit if proportional to 19% or \$16.6M if fully deferred; stake 19% (10-K, 2026 10-Qs) against 20% (Q3 2025 10-Q); Leasing guidance \$575M (Feb 2026) cut to \$475M (Jul 2026), a \$575M reading of the July release found to be a mis-read	S10, Exhibit S.10

Unknowns carried in the Spine

The consolidated register of everything the research could not find is Part XII §94, carried verbatim from the reconciliation's sections D.1 and D.2 with the facts that later closed some entries. Nineteen clusters group the entries that ask one question. The five largest, by the number of entries they merge and by what the chain's arithmetic depends on, are these. K-02, appraiser values for CFM56 engines (five entries): no half-life base value, no -5B value, no full-life value and no mid-life aircraft value was reached, so every valuation example stands on IBA's two -7B market values. K-01, module-exchange pricing (three entries): no provider's exchange fee, FTAI's included, was found, so every per-module figure is a bound. K-17, segment revenue and Adjusted EBITDA by year (three entries): the 2023 Leasing lines, the Adjusted EBITDA reconciliation and net income by year were not returned. K-05, a stated PMA discount to list (three entries): neither Chromalloy's nor HEICO's discount was disclosed, so the saving from the alternative-parts lever cannot be sized. K-18, the capital tied up in module inventory (two entries): the inventory change and the investing lines of the cash-flow statement were not retrieved, so the mechanism of S12 is supported by one balance and not by the statement's own lines. Also open, and named here because S10 turns on them: the servicing fee's rate and base and whether a promote exists (K-09), and the elimination method of S10.

Sources

1. FTAI Aviation Ltd., Form 10-K for FY2025, filed 27 Feb 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026012940/ftai-20251231.htm>

2. FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Jun 2026. <https://www.sec.gov/Archives/edgar/data/0001590364/000162828026051412/ftai-20260630.htm>
3. FTAI Aviation Ltd., Form 10-Q for the quarter ended 31 Mar 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026029335/ftai-20260331.htm>
4. FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Sep 2025. <https://www.sec.gov/Archives/edgar/data/1590364/000159036425000041/ftai-20250930.htm>
5. FTAI Aviation Ltd., Form 10-Q for the quarter ended 30 Sep 2024. <https://www.sec.gov/Archives/edgar/data/1590364/000159036424000021/ftai-20240930.htm>
6. FTAI Aviation Ltd., Form 10-K for FY2022. <https://www.sec.gov/Archives/edgar/data/1590364/000159036423000007/ftai-20221231.htm>
7. FTAI Aviation Ltd., Form 8-K and Exhibit 99.1 (internalization), 28 May 2024. https://www.sec.gov/Archives/edgar/data/1590364/000114036124027545/ny20030051x1_8k.htm
8. FTAI Aviation Ltd., Form 8-K, Jan 2025 (Audit Committee review), and Form 8-K Exhibit 99.1, Feb 2025 (conclusion). https://www.sec.gov/Archives/edgar/data/1590364/000114036125001452/ef20041894_8k.htm ; https://www.sec.gov/Archives/edgar/data/1590364/000114036125005179/ef20043996_ex99-1.htm
9. FTAI Aviation Ltd., Q2 2026 earnings release, 8-K Exhibit 99.1, 29 Jul 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026050622/ftai6302026earningsrelease.htm>
10. FTAI Aviation Ltd., Q4 and FY2025 earnings release, 25 Feb 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026011685/ftai123125earningsrelease.htm>
11. FTAI Aviation Ltd., Q1 2026 earnings release, 29 Apr 2026. <https://www.sec.gov/Archives/edgar/data/1590364/000162828026028390/ftai3312026earningsrelease.htm>
12. FTAI Aviation Ltd., Q3 2025 earnings release, 27 Oct 2025. <https://www.sec.gov/Archives/edgar/data/1590364/000159036425000038/ftai93025earningsrelease.htm>
13. FTAI Aviation Ltd., Q2 2025 earnings release, GlobeNewswire, 29 Jul 2025. <https://www.globenewswire.com/news-release/2025/07/29/3123626/35538/en/>
14. FTAI Aviation Ltd., Q4 and FY2024 results and QuickTurn Europe agreement, GlobeNewswire, 26 Feb 2025. <https://www.globenewswire.com/news-release/2025/02/26/3033379/35538/en/>
15. FTAI Aviation Ltd., "FTAI's Strategic Capital Initiative Secures Commitment for \$2.5 Billion of Asset-Level Debt Financing", GlobeNewswire, 26 Feb 2025. <https://www.globenewswire.com/news-release/2025/02/26/3032687/35538/en/>
16. FTAI Aviation Ltd., "FTAI Aviation Launches Strategic Capital Initiative and Announces 2025 Financial Guidance", GlobeNewswire, 30 Dec 2024. <https://www.globenewswire.com/news-release/2024/12/30/3002893/35538/en/>

17. FTAI Aviation Ltd., "FTAI Aviation Closes the Acquisition of LMCES (Lockheed Martin Commercial Engine Solutions)", GlobeNewswire, 9 Sep 2024. <https://www.globenewswire.com/news-release/2024/09/09/2943296/0/en/>
18. FTAI Aviation Ltd., "Announces Closing of QuickTurn Europe Joint Venture", GlobeNewswire, 5 Jun 2025. <https://www.globenewswire.com/news-release/2025/06/05/3094782/0/en/>
19. FTAI Aviation Ltd., "Completes Fundraising for its Inaugural Strategic Capital Vehicle ... \$2.0 Billion", GlobeNewswire, 27 Oct 2025. <https://www.globenewswire.com/news-release/2025/10/27/3174981/35538/en/>
20. FTAI Aviation Ltd., "FTAI Closes \$2.0 Billion Warehouse Financing Facility for Strategic Capital's Second Investment Vehicle", GlobeNewswire, 17 Aug 2026. <https://www.globenewswire.com/news-release/2026/08/17/3345919/35538/en/>
21. FTAI Aviation Ltd., "FTAI Aviation Prices Inaugural Asset-Backed Securitization", 22 May 2026, via Finviz and AviTrader, 26 May 2026. <https://finviz.com/news/356600/ftai-aviation-prices-inaugural-asset-backed-securitization> ; <https://avitrader.com/2026/05/26/ftai-aviation-prices-inaugural-abs-deal/>
22. FTAI Aviation Ltd., "FTAI Acquires 27 Boeing 737-700 Aircraft from WestJet", GlobeNewswire, 28 Sep 2026. <https://www.globenewswire.com/news-release/2026/09/28/3369685/35538/en/>
23. FTAI Aviation Ltd., "FTAI Aviation Announces the Launch of FTAI Power", GlobeNewswire, 30 Dec 2025. <https://www.globenewswire.com/news-release/2025/12/30/3211297/35538/en/>
24. FTAI Aviation Ltd., "FTAI Announces \$1.465 Billion Gas Turbine Generator Set Order Through J&F Power Systems", GlobeNewswire, 22 Jul 2026. <https://www.globenewswire.com/news-release/2026/07/22/3331360/35538/en/>
25. FTAI Aviation Ltd., "Multi-Year Materials Agreement with CFM International", GlobeNewswire, 22 Jan 2026 (title and summary). <https://www.globenewswire.com/news-release/2026/01/22/3223741/35538/en/>
26. FTAI Aviation Ltd., "FTAI Aviation Announces CFO Transition", GlobeNewswire, 6 Mar 2026. <https://www.globenewswire.com/news-release/2026/03/06/3250943/0/en/>
27. FTAI Aviation Ltd., "\$500 Million Share Repurchase Program", GlobeNewswire, 15 Sep 2026 (title only). <https://www.globenewswire.com/news-release/2026/09/15/3362653/35538/en/>
28. FTAI Infrastructure Inc., Form 10-K for FY2024 (listing and spin-off history). <https://www.sec.gov/Archives/edgar/data/1899883/000189988325000007/ftai-20241231.htm>
29. AAR Corp / FTAI Aviation, "AAR and FTAI Aviation extend their exclusive Serviceable Engine Products agreement ... through 2030", PR Newswire, 27 Mar 2025 (body as carried

- by ePlaneAI). <https://www.prnewswire.com/news-releases/aar-and-ftai-aviation-extend-their-exclusive-serviceable-engine-products-agreement-providing-cfm56-engine-material-to-the-global-aviation-aftermarket-through-2030-302412621.html>
30. CFM International, "The CFM56 engine family", undated, accessed 2026-10-03. <https://www.cfmaeroengines.com/engines/cfm56>
 31. GE Aerospace, "1 billion flight hours ... 15,000th CFM56-7B engine", 2019. <https://www.geaerospace.com/news/articles/manufacturing-paris-airshow-people-product/1-billion-flight-hours-world-class-experience>
 32. ISTAT, "CFM International CFM56-5B/-7B", undated; ISTAT Jetrader, Jul/Aug 2010 (half-life definition). <https://www.istat.org/ISTAT-Online/ISTAT-Online/Aircraft-Supporting-Assets/ArtMID/1205/ArticleID/1677/CFM-International-CFM56-5B-7B> ; <https://www.nxtbook.com/nxtbooks/naylor/ISTS0410/index.php?startid=16>
 33. Aircraft Commerce, Issue 34 (2004), Issue 58 (2008) and Issue 120 (2018). <https://aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs/Maintenance/2004/ISSUE%2034-MTCE.pdf> ; https://www.aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs1/Aircraft%20guides/CFM56-7B/ISSUE58_CFM56_7B_GUIDE.pdf ; https://www.aircraft-commerce.com/sample_article_folder/120_MTCE_B.pdf
 34. Aircraft Commerce, Maintenance & Engineering, Issue 27, 2003 (EGT margin). <https://aircraft-commerce.com/wp-content/uploads/aircraft-commerce-docs1/Maintenance/2003/ISSUE%2027-MTCE-A.pdf>
 35. AJW Group, ESN 889979 CFM56-7B26 mini-pack, 2022; StandardAero, ESN 892820 mini-pack, Mar 2025. https://www.ajw-group.com/storage/downloads/1644847608_esn_889979_mini_pack_cfm56-7b26.pdf ; <https://standardaero.com/wp-content/uploads/2025/03/ESN-892820-Mini-Pack-PDF.pdf>
 36. Leeham News, Bjorn's Corner, "Engine maintenance", 8 Mar 2024. <https://leehamnews.com/2024/03/08/bjorn-s-corner-new-aircraft-technologies-part-49-engine-maintenance/>
 37. Aircraft Value News, "Engine life limited parts pricing continues to rise: \$4m for CFM56-7", Nov 2019; "Concept of half to full life fluid ...", undated. <https://www.aircraftvaluenews.com/engine-life-limited-parts-pricing-continues-to-rise-4m-for-cfm56-7> ; <https://www.aircraftvaluenews.com/concept-of-half-to-full-life-fluid-as-aircraft-move-pass-mid-life/>
 38. MyAirTrade, engine status resource (LLP cost \$5.7M, June 2025). <https://www.myairtrade.com/resources/enginestatus>
 39. Safe Fly Aviation (consultancy blog, 2026): "Engine Shop Visit Costs Worldwide 2026"; "What determines aircraft engine overhaul costs?"; "CFM56-7B engine availability report

- 2026"; "CFM56 Engine Market Report 2026". <https://safefly.aero/?p=15606> ; <https://safefly.aero/blog-aircraft-engine-overhaul-cost-drivers/> ; <https://safefly.aero/cfm56-7b-engine-availability-report/> ; <https://safefly.aero/cfm56-engine-market-report-2026/>
40. Air Cargo Week, "The true cost of engine maintenance", 2025/26. <https://aircargoweek.com/the-true-cost-of-engine-maintenance/>
 41. Aviation Business News / Aviation Maintenance magazine (AVM), "Inside the engine MRO supply chain", 2025; Aviation Business News, "CFM56 turbofan aircraft engine", Sept 2025. <https://www.aviationbusinessnews.com/mro/mro-interviews-comments-articles/inside-the-engine-mro-supply-chain-why-repair-delays-are-rising-and-whats-driving-them/> ; <https://www.aviationbusinessnews.com/low-cost/cfm56-turbofan-aircraft-engine/>
 42. GE Aerospace, Bernstein Strategic Decisions Conference presentation, 27 May 2026. https://www.geaerospace.com/sites/default/files/geaerospace_bernstein_strategic_decisions_conference_presentation_052726.pdf
 43. GE Aerospace Investor Relations, "Recent events: your questions answered", 2026. <https://www.geaerospace.com/news/investor-relations/ir-updates/recent-events-your-questions-answered>
 44. General Electric Co. (GE Aerospace), Form 10-K for FY2025. <https://www.sec.gov/Archives/edgar/data/40545/000004054526000008/ge-20251231.htm>
 45. Bain & Company, "Get a step ahead of the engine maintenance capacity crunch", 2024. https://bain.com/globalassets/noindex/2024/bain_brief_get_a_step_ahead_of_the_engine_maintenance_capacity_crunch.pdf
 46. Oliver Wyman, "The new MRO supply paradigm", Apr 2026. <https://www.oliverwyman.com/our-expertise/insights/2026/apr/aviation-mro-labor-and-material-supply-chain-paradigm.html>
 47. IBA, "Engine Values Release September 2025 (2025B)"; "Engine and Lease Rate Update H1 2026"; "It's a 'Lessors' Market", Apr 2024, via AviTrader, 25 Apr 2024. <https://www.iba.aero/resources/articles/engine-values-2025b-release-september-2025/> ; <https://www.iba.aero/resources/articles/iba-engine-and-lease-rate-update-h1-2026/> ; <https://avitrader.com/2024/04/25/iba-says-its-a-lessors-market/>
 48. India Seatrade News, "IBA warns of oversupply of narrowbody conversions and lease rate fall", 6 Sep 2026 (IBA's July 2025 rents). <https://indiaseatradenews.com/iba-warns-of-oversupply-of-narrowbody-conversions-and-lease-rate-fall/>
 49. Bloomberg via AJOT, "Used jets rents surge 44% as airlines pay up to buoy fleets"; Key.Aero, "Lease trends" (IBA research). <https://www.ajot.com/news/used-jets-rents-surge-44-as-airlines-pay-up-to-buoy-fleets> ; <https://www.key.aero/article/lease-trends>

50. Aviation Week: "CFM56 Overhaul Demand Remains Strong, GE Aerospace Says", 2026; "Safran Sees Leap Aftermarket Emerging By Mid-Decade", c. 2023/24; "CFM Aftermarket Shifting To Long-term Agreements, Safran Says", c. 2023/24; "Leap Aftermarket Poised For Growth", undated; "Parts Price Hikes Help Boost Safran, GE Aftermarket Sales", late 2023; "Pratt & Whitney Sees High-Single-Digit MRO Growth This Year", 2026; "Data Tool: The Future Of IAE V2500 Engine", 2023; "MRO Memo: A Seller's Market For Used Parts", undated; "MRO Memo: FTAI Pitches CFM56 To Power AI Boom", early 2026. <https://aviationweek.com/mro/aircraft-propulsion/cfm56-overhaul-demand-remains-strong-ge-aerospace-says> ; <https://ngtest.aviationweek.com/mro/aircraft-propulsion/safran-sees-leap-aftermarket-emerging-mid-decade> ; <https://aviationweek.com/mro/aircraft-propulsion/cfm-aftermarket-shifting-long-term-agreements-safran-says> ; <https://aviationweek.com/mro/supply-chain/leap-aftermarket-poised-growth> ; <https://ngtest.aviationweek.com/mro/aircraft-propulsion/parts-price-hikes-help-boost-safran-ge-aftermarket-sales> ; <https://aviationweek.com/mro/aircraft-propulsion/pratt-whitney-sees-high-single-digit-mro-growth-year> ; <https://ngstage.aviationweek.com/mro/aircraft-propulsion/data-tool-future-iae-v2500-engine> ; <https://aviationweek.com/mro/workforce-training/mro-memo-sellers-market-used-parts> ; <https://aviationweek.com/mro/emerging-technologies/mro-memo-ftai-pitches-cfm56-power-ai-boom>
51. ePlaneAI, "Forecast for US commercial fleet and MRO growth through 2026" (Aviation Week 2026 Fleet & MRO Forecast figures); "CFM56 engine values highlight investor opportunities and risks". <https://www.eplaneai.com/de/news/forecast-for-us-commercial-fleet-and-mro-growth-through-2026> ; <https://www.eplaneai.com/news/cfm56-engine-values-highlight-investor-opportunities-and-risks>
52. Leeham News, "GE Aerospace FY and Q4 2025 Earnings ...", 22 Jan 2026; "RTX 2025 Earnings ...", 27 Jan 2026. <https://leehamnews.com/2026/01/22/ge-aerospace-fy-and-q4-2025-earnings-thrust-higher-propelled-by-services-growth-leap-volume-and-expanding-margins/> ; <https://leehamnews.com/2026/01/27/rtx-2025-earnings-commercial-aerospace-leads-growth-as-pratt-advances-gtf-recovery/>
53. Safran, FY2025 Results & Investor Update presentation, 13 Feb 2026. <https://www.safran-group.com/download/media/450393>
54. IATA Maintenance Cost Conference, Aerodynamic Advisory, "Retirements, part-outs, dismantling", 2023; AviTrader, "Global airline industry set for record growth in 2024" (IBA data), 24 Sep 2024; Informa Markets, Commercial Forecast story, page 5 (Cirium figures). https://www.iata.org/contentassets/3f8981eb437e4e16808639bc9d19d5c7/mcc202_day02_0915-0945_retirements-partouts-dismantling_aerodynamic_murby.pdf ; <https://avitrader.com/2024/09/24/global-airline-industry-set-for-record-growth-in-2024> ; <https://informamarkets.turtl.co/story/com-forecast/page/5>

55. The Flying Engineer, "A320neo grounded 2026: Pratt & Whitney crisis" (citing Cirium). <https://theflyingengineer.com/a320neo-grounded-2026-pratt-whitney-crisis/>
56. Visual Approach, "Wave of early LEAP-1A/1B shop visits frustrating operators", undated. <https://visualapproach.io/wave-of-early-leap-1a-1b-shop-visits-frustrating-operators/>
57. CFM International / Safran / GE Aerospace, "CFM secures certification for LEAP-1B durability upgrades", 18 Jul 2026. <https://www.cfmaeroengines.com/press-articles/cfm-secures-certification-for-leap-1b-durability-upgrades>
58. TravelPulse, "FAA Raises Boeing 737 MAX Production Cap", Oct 2025; CAPA, "Boeing testing production rate of 47 aircraft per month ...", May 2026. <https://www.travelpulse.com/news/airlines-airports/faa-allows-boeing-to-build-42-737-max-planes-per-month> ; <https://centreforaviation.com/news/boeing-testing-production-rate-of-47-aircraft-per-month-cautious-of-increasing-to-52-ceo-1360867>
59. Argus Media, "Airbus targets 820 aircraft deliveries in 2025"; Air Data News, "Airbus targets around 870 deliveries in 2026 ...". <https://www.argusmedia.com/de/news-and-insights/latest-market-news/2659923-airbus-targets-820-aircraft-deliveries-in-2025> ; <https://www.airdatanews.com/airbus-targets-around-870-deliveries-in-2026-as-engine-issues-slow-a320neo-ramp-up>
60. Wikipedia, "List of Boeing 737 MAX orders and deliveries", 2026. https://en.wikipedia.org/wiki/List_of_Boeing_737_MAX_orders_and_deliveries
61. Air Lease Corp, Form 10-K for FY2025. <https://www.sec.gov/Archives/edgar/data/1487712/000162828026007707/al-20251231.htm>
62. AerCap Holdings N.V., Form 20-F for FY2013 (maintenance rents and end-of-lease compensation definitions, via Part II). <https://www.sec.gov/cgi-bin/browse-edgar?action=getcompany&CIK=0001378789&type=20-F>
63. AerSale Corp., FY2025 results release, 5 Mar 2026; Form 10-K for FY2025. <https://www.sec.gov/Archives/edgar/data/1754170/000110465926024101/asle-20260305xex99d1.htm> ; <https://www.sec.gov/Archives/edgar/data/1754170/000110465926025574/asle-20251231x10k.htm>
64. HEICO Corporation, Form 10-K for the fiscal year ended 31 Oct 2025. <https://www.sec.gov/Archives/edgar/data/46619/000004661925000082/hei-20251031.htm>
65. Chromalloy, "Chromalloy Secures FAA Approval of CFM56 High Pressure Turbine Blade PMA", 30 Oct 2025. <https://www.chromalloy.com/chromalloy-secures-faa-approval-of-cfm56-high-pressure-turbine-blade-pma/>
66. eCFR, 14 CFR Part 21 Subpart K; FAA Order 8110.37F (via Part VI). <https://www.ecfr.gov/current/title-14/chapter-I/subchapter-C/part-21/subpart-K>
67. In Practise, "FTAI, Chromalloy and CFM56 HPT Blade PMA" (expert interview digest), undated. <https://inpractise.com/articles/ftai-chromalloy-and-cfm56-hpt-blade-pma>

68. Komodo Capital (Substack), "Why FTAI Aviation (\$FTAI) Presents a Rare Opportunity"; Kairos Research (Substack), "Why I bought FTAI Aviation" (investor blogs). <https://komodocapital.substack.com/p/why-ftai-aviation-ftai-will-fly-in> ; <https://kairosresearch.substack.com/p/why-i-bought-ftai-aviation>
69. Insider Monkey, FTAI Q3 2025 earnings call transcript, Oct 2025. <https://www.insidermonkey.com/blog/ftai-aviation-ltd-nasdaqftai-q3-2025-earnings-call-transcript-1635931/>
70. Motley Fool, FTAI Q4 2025 earnings call transcript, 26 Feb 2026; MarketBeat, Q4 2025 call highlights, 26 Feb 2026. <https://www.fool.com/earnings/call-transcripts/2026/02/26/ftai-ftai-q4-2025-earnings-call-transcript/> ; <https://www.marketbeat.com/instant-alerts/ftai-aviation-q4-earnings-call-highlights-2026-02-26/>
71. GuruFocus, FTAI Q2 2026 earnings call transcript, 30 Jul 2026; Quartr, FTAI Q2 2026 summary; Nasdaq.com and MarketBeat coverage of the same call (URLs not captured). <https://www.gurufocus.com/stock/FTAIN.PFD/transcripts/8991621> ; https://quartr.com/events/ftai-aviation-ltd-ftai-q2-2026_ozdaN73d
72. TheValueist (X), summary of the FTAI Q4 2025 earnings call, 26 Feb 2026 (call coverage). <https://x.com/TheValueist/status/2027078155635503303>
73. FilingReader, "Yantai Jereh subsidiary secures \$1.47bn turbine contract", 22 Jul 2026. <https://filingreader.com/news-wire/shenzhen/2026-07-22/yantai-jereh-subsi-dary-secures-147bn-turbine-contract>
74. International Energy Agency, "Energy and AI", executive summary, Apr 2025. <https://www.iea.org/reports/energy-and-ai/executive-summary>
75. Power Engineering, "Data centers drive record surge in GE Vernova power equipment orders ...", Jul 2026; Oilprice, "The Gas Turbine Shortage Just Became AI's Biggest Constraint", 2026. <https://www.power-eng.com/gas/turbines/data-centers-drive-record-surge-in-ge-vernova-power-equipment-orders-as-turbine-slots-tighten-through-2030/> ; <https://oilprice.com/Energy/Energy-General/The-Gas-Turbine-Shortage-Just-Became-AIs-Biggest-Constraint.amp.html>
76. Data Centre Magazine, "Can Old Aeroplane Engines Powering Data Centres Take Off?", undated (bridging power). <https://datacentremagazine.com/news/can-old-aeroplane-engines-power-data-centres-take-off>
77. StandardAero, Inc., Form 10-K for FY2025; MTU Aero Engines, FY2025 results, 24 Feb 2026; Lufthansa Technik, FY2025 results. <https://www.sec.gov/Archives/edgar/data/2025410/000119312526072618/saro-20251231.htm> ; <https://www.mtu.de/newsroom/press/latest-press-releases/press-release-detail/figures-for-2025-mtu-stays-on-course-for-growth/> ; <https://www.lufthansa-technik.com/en/lufthansa-technik-stable-on-course-for-growth-2601a780f4c4031c>

78. The Carlyle Group Inc., Form 10-K for FY2025; Willis Lease Finance Corporation, Forms 10-K for FY2019 and FY2020 (vehicle fee scales). <https://www.sec.gov/Archives/edgar/data/1527166/000152716626000009/cg-20251231.htm> ; <https://www.sec.gov/Archives/edgar/data/1018164/000101816421000013/wlfc-20201231.htm>
79. RTX / Pratt & Whitney, "IAE AG and FTAI Aviation Sign Strategic V2500 Engine Maintenance Services Agreement", 6 Jun 2024. <https://www.rtx.com/prattwhitney/newsroom/news/2024/06/06/iae-ag-and-ftai-aviation-sign-strategic-v2500-engine-maintenance-services-agreem>
80. Benzinga, "What's Going On With FTAI Aviation Stock Today", 15 Jan 2025; StockStory via Wedbush, 20 Feb 2025; law-firm case pages on S.D.N.Y. No. 25-cv-00541 (ktmc.com, rgrdlaw.com; via the orchestrator notes, URLs not captured). <https://benzinga.com/25/01/43103333/whats-going-on-with-ftai-aviation-stock-today> ; <https://investor.wedbush.com/wedbush/article/stockstory-2025-2-20-why-ftai-aviation-ftai-stock-is-trading-up-today>
81. Research inputs cited for derived figures and definitions: /home/user/SPOT/ftai-primer/research/reconciliation.md (section E.2 binding numbers; tensions R-001 to R-070; clusters K-01 to K-19), /home/user/SPOT/ftai-primer/research/orchestrator-notes.md (resolutions and the May 2026 securitization) and /home/user/SPOT/ftai-primer/research/term-registry.md (section 4, the Spine vocabulary), 2026-10-03.